

AI-Empowered Cooperative One-Surgeon-Four-Arm Robotic System for Solo-Hysterectomy

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Abstract

The increasing robotic and intelligent techniques are enhancing surgeon's operational capabilities for finer surgical intervention, yet AI-enabled surgical robots for solo laparoscopic procedures remain nascent and clinically unvalidated. Here, we present an AI-empowered one-surgeon-four-arm robotic system that enables one surgeon to perform a complete hysterectomy (79mins) independently for the first time. The system integrates two intelligent robots that autonomously interpret surgical scenes to optimize control strategies and adapt intraoperative views on-the-fly. We further introduce multiple levels of cooperative control (LoCC) paradigm that is clinically feasible for surgical autonomy with explicit surgeon-AI-robots role definitions, facilitating automation of a solo hysterectomy in operating theaters. Phantom and cadaver experiments demonstrate our system's superior performance in visual parsing, surgical scene understanding, and cooperative control, which enables efficient and highly autonomous coordination (>55mins full-automation) of four-arm surgical tasks, demonstrating our LoCC framework's scalability for long-horizon task automation in various laparoscopic treatments through a human-robot cooperative paradigm.

Keywords: Medical Robotics; Robotic-assisted Surgery; Human-in-the-loop Control; Robotic Hysterectomy

Introduction

Minimally invasive surgery (MIS) has revolutionized the medical field in recent decades, as it offers the benefits of less invasiveness and accelerated recovery time for patients compared with open surgery [1, 2]. MIS typically necessitates a collaborative effort between the primary surgeon and several assistants responsible for essential surgical sub-tasks (e.g., holding the laparoscope or manipulating tissues). However, such tedious tasks demand frequent correction of commands from surgeons that guide assistants to achieve their desired targets, disrupting the precision of surgical actions and the stability of assistants' operations [3]. On top of this, the collaborative nature of the multi-person surgical paradigm also limits the feasibility of conducting surgeries under extreme conditions, such as in rural places with limited clinical resources, or under night shifts and military zones where surgical personnel are scared, etc. The concept of performing complex MIS with more than two biological arms assisted by AI-driven autonomous capabilities remains in its fantasy and currently exists only as speculative visions of future technology depicted in fiction films. Yet, enabling modern surgery with such capability by surgical robots holds the potential to reshape current clinical workflows. In aforementioned real-world and extreme scenarios, empowering a primary surgeon to collaborate with autonomous robots can dramatically enhance responsiveness, safety, and surgical accessibility. Looking forward, these capabilities may also support future medical needs in confined or isolated settings, such as polar research

stations, deep-sea habitats, and space-based missions, where traditional multi-person surgical teams are logistically impractical.

To empower surgeons with such capability to independently perform complex surgeries, extra surgical robots functioning as supplementary "biological arms" are critically needed. With sophisticated designs and creations, these robots are expected to intelligently support long-horizon and delicate manipulations by mastering the low-level control tasks (e.g., the laparoscope control or manipulating tissues), while surgeons can continuously focus on the complicated primitive operations, such as ligament dissection, uterine artery sealing, wound anastomosis, etc. By incorporating a human-in-the-loop interaction mechanism, the overall robotic system can intelligently cooperate with surgeons and adapt to the dynamic surgical scenarios [4], realizing a more efficient and autonomous paradigm of robotic surgery. However, in such variable environments requiring more than two arms, off-the-shelf teleoperation systems (e.g., da Vinci system [5]) are significantly limited by the absence of additional intelligent and cooperative robotic arms [6, 7]. When referring to laparoscopic surgery, the difficulty of intelligent manipulation is further increased owing to challenges such as dexterous manipulation in tight cavities [8], eye-hand coordination [9, 10, 11], and surgical safety [12, 13] during complex operating procedures. Realizing a higher level of intelligent human-robot cooperative solo surgery, particularly for complicated and prolonged tasks like laparoscopy, requires that robots reliably perform standard tasks autonomously. Automating these fundamental procedures not only improves surgical efficiency but also serves as a critical step-

ping stone toward a higher level of autonomy in robotic-assisted operations.

Current laparoscopic hysterectomy demands a high level of cooperation between the surgeon and assistants, where a specific assistant needs to maneuver and hold the uterus accordingly to comply with the main operation (i.e., dividing ligament and peritoneum, uterine vessels and ligament, and transecting the vagina) from the surgeon, and another assistant serves as a laparoscope holder, providing surgeons with proper *in-vivo* views. Hence, achieving solo-hysterectomy is a suitable track to showcase the superiority of a higher level of robot-assisted autonomy in surgical operations. However, critical challenges to achieving solo-hysterectomy lie in the versatile robotic designs for dexterous uterine manipulation, effective control of the laparoscopic field of view (FoV) aligned with surgeons' preferences, and the artificial intelligence (AI)-driven automation, such as instrument tracking, surgical phase recognition, as well as intuitive and low learning-curved interactions for surgeons, such as advanced strategies to balance the ternary relation of surgeon-AI-robots role [14].

The delineation of roles among the surgeon-AI-robots in the execution of a surgical procedure is vital, yet prior landmark works primarily emphasized the importance of AI-robots and introduced five levels of autonomy (LoA) [15, 16]. At LoA 0 (no autonomy), where the robot moves according to the user's command, e.g., teleoperation in the da Vinci surgical system [5], whereas at LoA 1 (robot assistance), the robot provides certain mechanical guidance or assistance during surgical tasks [17]. For higher autonomy levels, AI is gradually incorporated to guide robots performing surgical tasks. In LoA 2 (task autonomy), the specified task initialized by the surgeon can be finished by robots, e.g., debridement [18, 19], suturing [8, 20], and blood suction [21], while in LoA 3 (conditional autonomy), surgeons make decisions on strategies generated by robotic systems. However, LoA 4 (high autonomy) and LoA 5 (full autonomy) require a robot to perform an entire surgery with/without the need from a human, yet off-the-shelf technologies and surgical robots cannot satisfy such paradigms. To the best of our knowledge, no previous work addressed the collaborative intelligence between robots and surgeons throughout an entire surgical procedure, in which robots autonomously act as independent agents and cooperate with the surgeon's main operation, boosting the completion of a long-horizon surgical procedure (e.g., laparoscopic hysterectomy). The central limitation of existing approaches is their lack of in-depth investigations of the entire surgical procedure, often neglecting the critical and indispensable role of the surgeon's expertise, which AI has yet to replicate [22]. To this end, it is essential to integrate the complementary strengths of the surgeon, AI, and robots throughout the entire surgical procedure, which requires a rethinking and optimization of the interaction paradigms within our proposed system, realizing a seamless surgeon-AI-robots cooperative hysterectomy.

Despite the off-the-shelf studies and attempts on robot-assisted automatic surgeries [22], experimental validations with complete technical frameworks supporting solo surgeries in on-site operating theaters or pre-clinical settings remain scarce. Moreover, the state-of-the-art works lack comprehensive assessments of autonomous/semi-autonomous AI-empowered surgical task performance, which should encompass diverse robotic operations from laparoscope to uterus manipulator, prolonged manipulations in dynamic surgical contexts, varying levels of surgeon-AI-robots interactions, and tests from phantom to ca-

da-ver trials. Additionally, achieving real-time online AI for cooperative robotic control that is precisely aligned with surgeons' actions, integrated seamlessly into clinical workflows, and meets clinical demands remains a complex and unresolved challenge [23]. All these factors would introduce obstacles to building a reliable surgical AI and should be clinically tested, avoiding surgical misoperation and guaranteeing surgical safety [24]. In this regard, systematic experiments, even human cadaver studies, are necessary to verify the surgeon-AI-robots models for actual clinical applications. Although some works have explored the possibility of using AI models to assist with knot tying [25] or suturing [26], they are limited to leveraging AI models for surgical sub-tasks, rarely considering running through the whole solo surgery in practice. For the advancement of the clinical value of AI-robots models, experimental results in real clinical settings are frequently suggested in smart healthcare-related guidelines [2]. To date, no reported work has been done on validating the surgeon-AI-robots to assist solo surgery in human pre-clinical settings for hysterectomy.

In this study, we develop a novel AI/robot-enabled one-surgeon-four-arm (OSFA) system (Fig. 1A), enabling one single surgeon to perform a solo robotic hysterectomy with two robots dynamically cooperating within the whole procedure. For the robot design, we built our surgical platform targeting clinical deployment, consisting of a robotic laparoscope (Fig. 1B) and uterine manipulators (Fig. 1C) acting as extra "biological arms" for surgeons. Beyond our previous work, we further combined the robotic laparoscope and developed a novel interaction scheme in real-time (Fig. 1D), which enables surgeons to perform hysterectomy more efficiently and cooperatively. For the AI aspect, we consider the instrument as a salient object to focus on, and then we introduce a long-term memory-based algorithm for instrument tracking during long-horizon dynamic surgical procedures, serving as core cues for surgeon-AI-robots interaction, along with utilizing phase recognition to achieve overall understanding of surgical scenes (Fig. 1E). The proposed OSFA system is highly automated and allows the surgeon to make high-level decisions on both the laparoscope and uterine control via a cooperative paradigm. Our system does not require extra sensing devices and keeps consistent settings with the clinical requirements, which is convenient for surgeons to adapt and collaborate, endowing nonverbal, feet-off, and intelligent controls of the laparoscope and uterus (Fig. 1F and G) for a delicate operation. We further validate the overall system by a 30-year-experienced surgeon to perform the hysterectomy task while the system runs semi-automatically. To the best of our knowledge, this is the first work that demonstrates a highly automated surgical robotic system cooperates with the surgeon to complete a solo-hysterectomy, paving a surgeon-AI-robots cooperative and scalable paradigm for realizing a higher level of task autonomy under long-horizon surgical operations.

Results

In this section, we first present the design results of our OSFA system, showcasing the developed system architecture and the cooperative control workflow for the intelligent hysterectomy. We then evaluate the accuracy of the maskless instrument tracking, which enables robust surgeon-robots interaction. A comparative analysis is conducted between manual assistance and OSFA-enabled autonomous assistance in a phantom study involving a representative hysterectomy procedure. Finally, we report results from a female cadaver study, where the surgeon

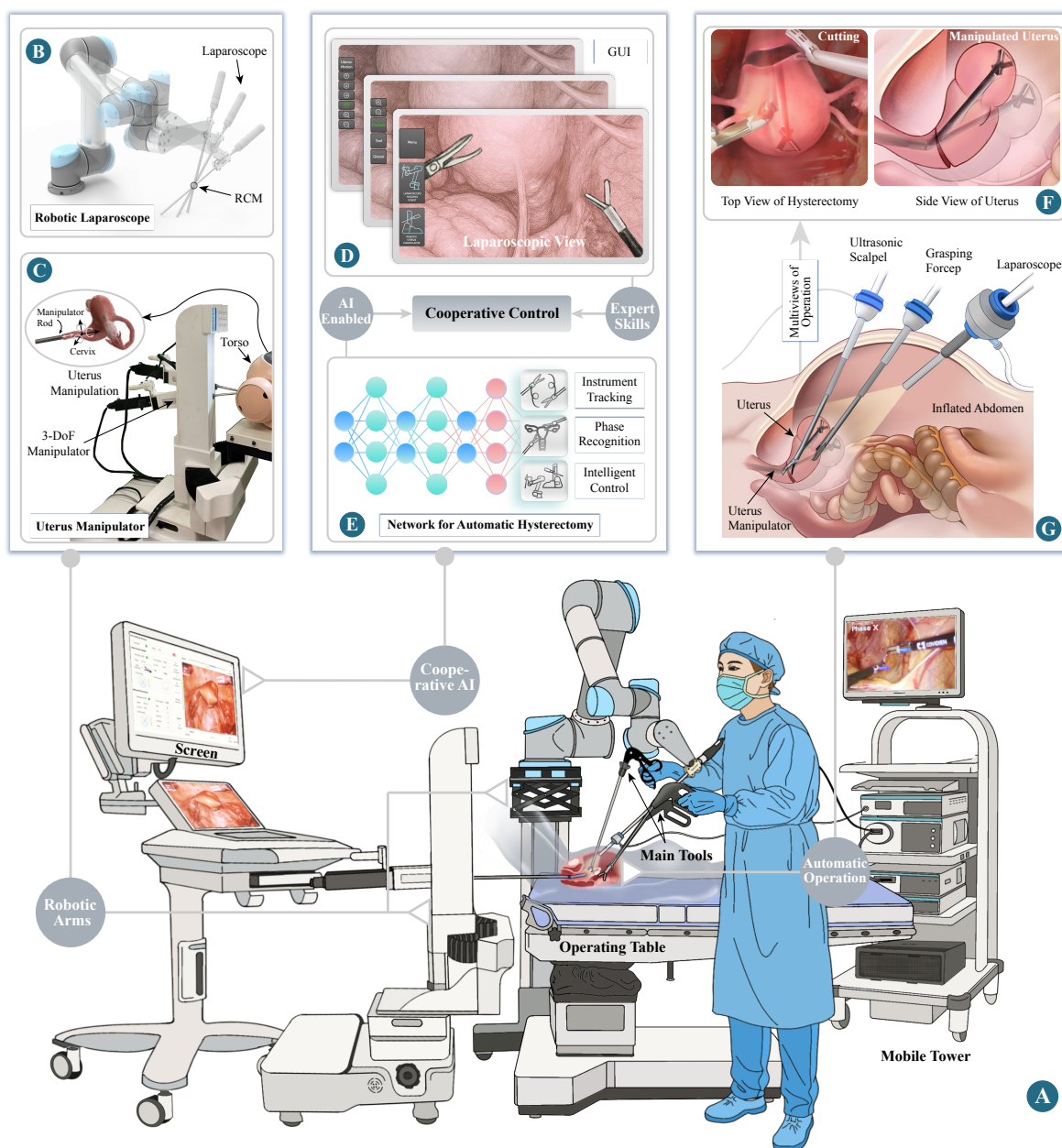


Figure 1 Our one-surgeon-four-arm system for conducting intelligent and cooperative robotic hysterectomy. (A) The overview of our OSFA system, in which the surgeon manipulates the two main surgical tools with two robots intelligently controls the laparoscope and uterus manipulator, providing optimal laparoscopic view and exposing the uterus for a cutting operation. (B) Robotic arm one: robotic laparoscope with RCM constraint. (C) Robotic arm two: uterus manipulator with a 3-DOF tool that drags the uterus within the abdomen. (D) Human-robot interactive interface. The surgeon can switch to the manual mode and utilize the tool to click the button on the interface, optimizing the motion of the laparoscope and the uterus manipulator. With AI networks, we can formulate our intelligent cooperative control with a high level of surgical autonomy, balancing operational efficiency and accuracy. (E) AI network for online surgical instrument tracking, operating phase recognition, and intelligent robotic controls in hysterectomy. (F) Robotic laparoscopic view of the bi-manual operation of the uterus cutting by two surgeon-controlled main surgical tools, with the side view of the uterus motion by the robotic manipulator. (G) Four-arm collaborated minimally invasive hysterectomy within the inflated abdomen.

successfully completed the hysterectomy solely using the OSFA system, demonstrating its remarkable capability for enhanced autonomous and efficient surgery.

Integrated hardware for enhanced robotic hysterectomy

Robotic laparoscope. We evaluate the robotic laparoscope with respect to three critical intraoperative requirements from the surgeon's perspective: (1) compliance with RCM constraint, (2) avoidance of interference with manual surgeon operations, and (3) ease of installation and removal during surgery. Experimental validation confirmed that the laparoscope provided stable 4-DOF motion (i.e., pitch/yaw/insertion/rotation) under the

RCM constraint using a soft-RCM control scheme. As shown in Fig. 2E2, the laparoscope maintained precise RCM-based motion throughout the procedure, with an RCM error below 0.1mm . To minimize workspace interference, the designed linkage—based on an equilateral triangle configuration (ETC)—successfully elevated the robot wrist link away from the patient by 31.9cm while preserving the consistent workspace (Fig. 2E1) to our prior work [17]. As shown in Fig. 2E3, the measured minimum distance from the robot wrist link to the RCM plane is 17.2cm , ensuring sufficient clearance during operation. Furthermore, the quick-release mechanism facilitates tool changes within seconds, enabling seamless laparoscope removal for tasks such as

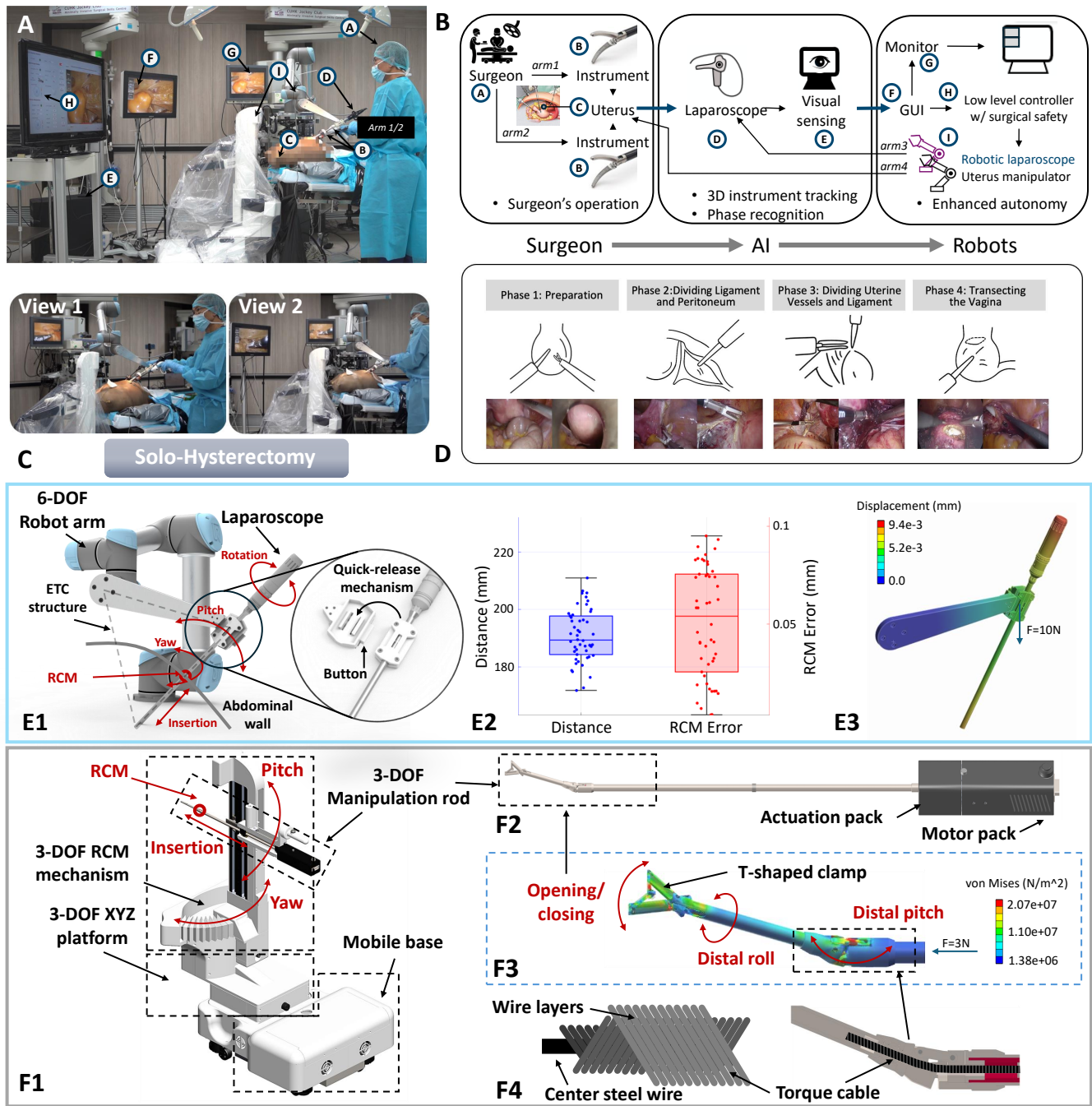


Figure 2 Schematic overview and hardware design of our one-surgeone-four-arm system for total laparoscopic hysterectomy. (A) Overview of the OSFA system setup, featuring a robotic arm for laparoscope control and a robotic manipulator for uterus manipulation. Both robots can be operated via a GUI panel overlaid on the image screen. (B) System control architecture illustrating the intelligent interaction strategy between the surgeon and robots in the OSFA system. (C) Multi-view illustrations of the cooperative solo-hysterectomy performed by the surgeon in collaboration with AI and robotic systems during the cadaver experiment. (D) Illustration and representative images from clinical videos showing four key phases of the total laparoscopic hysterectomy procedure. (E) Mechanical design of the robotic laparoscope. (E1) The robotic laparoscope is composed of a 6-DOF UR5 robotic arm and a stereo laparoscope. A quick-release structure is designed for efficient detachment and installation. (E2) Interval distance (left) between the robot wrist link to the RCM plane, and RCM error (right) during the cadaver study. (E3) Deformation simulation of the binding mechanism. (F) Mechanical design of the robotic uterus manipulator. (F1) The manipulator includes a 3-DOF RCM mechanism, a 3-DOF XYZ platform, a 3-DOF manipulation rod, and a mobile base. (F2) Design of the uterus manipulation rod, which consists of a distal mechanism, actuation pack, and motor pack. (F3) Design of the distal mechanism of the manipulation rod and stress analysis. (F4) Illustration of the torque cable structure embedded in the distal mechanism.

lens cleaning or repositioning (Fig. 2E1). Force-displacement testing (Fig. 2E3) demonstrated high mechanical stability, with camera displacement remaining a maximum 0.01 mm under a 10 N load, indicating sufficient rigidity during dynamic operation. These results validate that the system satisfies essential functional demands for intraoperative use.

Robotic uterus manipulator. We have developed a dedicated robotic uterine manipulation system, as shown in Fig. 2. This robot is positioned at the vaginal end to replace the assistant in manipulating the uterus during surgery, providing the primary surgeon with sufficient space to operate around the uterus. The robot features a mechanical Remote Center of Motion (RCM) design that allows the uterus manipulator to perform pitch, yaw,

and insertion movements around the cervix, as depicted in Fig. 2-F1. Specifically, the pitch ranges from -50° to $+34^\circ$, the yaw ranges from -37° to $+37^\circ$, and the insertion ranges from 0 mm to 30 mm, ensuring compatibility with the human anatomical structure and minimizing tissue strain during manipulation. Yaw motion is facilitated by a ring gear mechanism with the axis passing through the RCM point, while pitch motion is achieved through a novel Single-Motor Bilinear-Guided (SMBG) mechanism, which provides a wide range of pitch angles within a compact design suitable for placement between the patient's legs. The accuracy of the RCM point was evaluated through four sets of experiments, and the mean absolute error (MAE) of the RCM point is 0.373 mm, which is considered clinically acceptable for the soft and flexible uterine tissue [27].

Furthermore, the robot features a motorized 3-DOF uterine manipulation rod that enables distal pitch, roll, and opening/closing movements, as shown in Fig. 2-F2 and F3. The rod consists of two main components: a motor pack housing three motors, and an actuation pack containing the drive mechanism and instrument tip, connected via flanges to form an integrated 3-DOF instrument. The instrument's distal pitch ranges from 0° to $+30^\circ$, jaw opening from 0° to $+60^\circ$, and distal roll from -45° to $+45^\circ$, providing enhanced flexibility in intrauterine manipulation without increasing vaginal compression. Fig. 2-F3 presents the stress analysis of the manipulation rod in simulation. When a force of 3N is applied to the rod to simulate the insertion motion, the maximum stress observed is $2.07\text{e}+7$ Pa, which is below the yield strength of $2\text{e}+8$ Pa, demonstrating the structural stability of the design. The T-shaped tip of the rod reduces uterine compression, while a torque cable enables distal roll and jaw control, as shown in Fig. 2-F4. The distal diameter of the rod was minimized to 6 mm to ensure broad applicability across a wide patient population, particularly accommodating older patients who typically have narrower cervical canals [28]. The independent joint operation, combined with current limitations to prevent excessive force, ensures patient safety. In summary, the robot's mechanical RCM mechanism, elongated SMBG structure, integrated multi-DOF uterine instrument, and joint force constraints effectively meet clinical requirements, significantly enhancing uterine manipulation in laparoscopic hysterectomy procedures.

Integrated software framework for enhanced robotic hysterectomy

Control framework and GUI design. Our OSFA system supports the control and interaction with two robots through various interfaces (Fig. 3A), and surgeons or assistants use these interfaces to send control signals to the robots. Simultaneously, surgeons receive visual feedback from the laparoscope to determine whether the robots have reached the optimal positions. We have implemented a variety of methods for control and communication between the surgeon and the OSFA system, encompassing fundamental tele-operation through remote control and simple command inputs with the extra device or voice. We also developed an advanced AI-driven automatic control method that can determine the direction and position where the robot needs to move based on the current surgical scene.

Moreover, we established a Graphical User Interface (GUI)-based control scheme that is developed upon AI-driven automation and leverages the surgeon's domain-specific knowledge for direct control, enabling an intuitive cooperative control paradigm to the robotic system without additional external de-

vices for signal input. The GUI display for controlling the two robots is illustrated in Fig. 3B, consisting of three pages: the main page, "Menu", the uterine motion control page, "Uterus Motion", and the laparoscope motion control page, "Camera Motion". It is implemented through our inSTRument Mouse (STRUM)-based human-machine interface (HMI) method, which interprets instrument placement within a specific area as a click, and the primary surgeon can directly push the virtual buttons on the GUI with the surgical instruments they hold, facilitating a human-in-the-loop interaction with the system.

Specifically, when the surgeon places the instrument in a designated location and holds it there for a moment, the "Menu" page will unfold, showing that the surgeon has successfully entered the GUI control mode. From this page, the operator can access the second layer of pages to control the two robots by placing the instrument tip into the corresponding box on the screen. The "Uterus Motion" page indicates the movement of the uterus in the laparoscopic FoV in six directions: "Up/Down", "Left/Right", and "Insertion/Retraction", which are achieved by the robot's motion along the pitch, yaw, and insertion DOF, respectively. The "Camera Motion" page allows the control of the laparoscope's movement in the "Zoom in/out" direction, switching between the main surgical instrument for automatic laparoscope tracking with "Tool", and "Global" for quickly returning to the global view state. The details of our control logic can be found in Table. S1.

Surgical instrument detection and tracking. The implementation of the GUI-based control requires the accurate detection and tracking of the operating instruments during surgery. To obtain precise instrument positions, we design a novel marker-free multi-instrument 3D tracking algorithm to predict the rotated bounding box of each instrument, empowering robust instrument tip detection and tracking within complex surgical environments. We first developed a Multi-Task Orientated Instrument network (MTOI-Net) based on YOLO architecture [29] by leveraging its excellent balance between computational efficiency and accuracy. As shown in Fig. 3C and Fig. S1, we collected the dataset from both phantom platform and real-world hysterectomy procedures, and then labeled a total of 11313 instances from 4404 images with different instruments of varied sizes (10218 instances from 3975 images for training and the rest for validation). Given video frames, the MTOI-Net predicted the instrument's category, orientated box, and insertion direction simultaneously. Then we introduced a Long-Term Memory-based Tracker (LTMT) to robustly track the aforementioned instruments online, with an adaptive stereo matching scheme to efficiently estimate the depth of the tracked instrument, providing spatial knowledge for downstream autonomous controls of the laparoscope. The details of our algorithms can be found in the following Methods section.

The tracking performance and the comparison with two representative tracking approaches, i.e., DeepSORT [30] (detection-based tracking) and PAWSS [31] (initialized region-based tracking) in the medical field are shown in Fig. 3D and E, respectively. The quantitative comparison results are summarized in Table. S2. We sampled an unseen 26-minute clinical video under dynamic environments and tested the overall tracking process. The PAWSS exhibited the lowest tracking accuracy of 3.8% as it lost tracking over time due to fast instrument motions and struggled to recover the tracking process. The DeepSORT achieved 84.3% accuracy as it can recover from tracking failures through re-detection, but suffered from 294 ID-switches due to occasion-

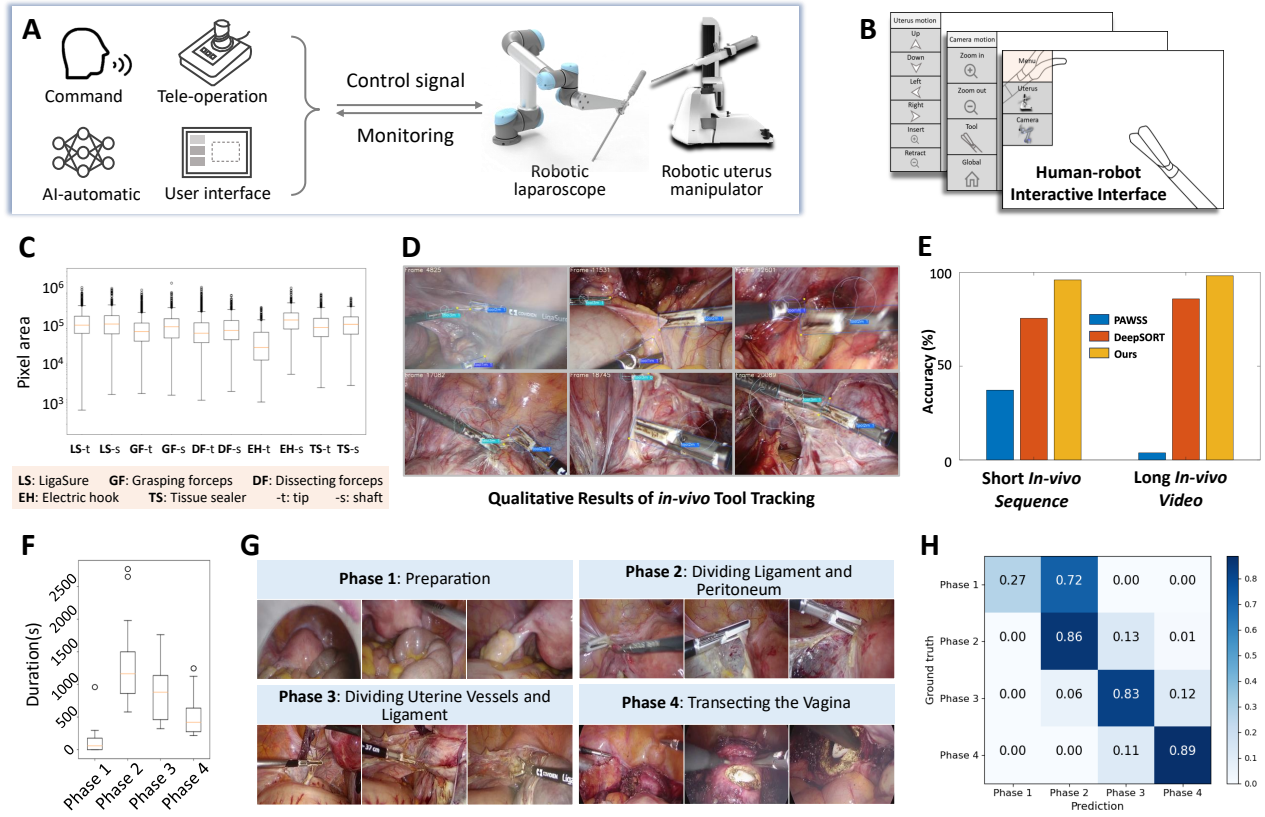


Figure 3 Illustration of the software design of the OSFA system, along with experimental data and results of surgical visual perception algorithms. (A) Control framework for the two robots in the OSFA system, supporting multiple interfaces for human-robot interaction. (B) Illustration of the designed GUI for surgeons to control the robots, with three pages: the main menu, uterus motion control, and laparoscope manipulation. (C) Data collection statistics from in-house surgical videos used for surgical instrument detection. (D) Sample snapshots from long-duration tracking in challenging scenarios, including rapid instrument and field-of-view movements, smoke, and blood contamination. (E) Quantitative comparison of instrument tracking performance under various conditions using PAWSS, DeepSORT, and our proposed method. (F) Statistical distribution of video durations across different key phases of the total laparoscopic hysterectomy procedure. (G) Example frames depicting the four surgical phases of total laparoscopic hysterectomy. (H) Confusion matrix showing the surgical phase recognition results across different phases.

ally incorrect detection caused by motion blur or the presence of similar instruments. Therefore, these tracking approaches were not reliable for our virtual click interaction-based control mechanism, and a more accurate continuous tracking approach is demanded. By contrast, our method achieved an impressive tracking accuracy of 98.6% and with only 4 ID-switches observed during the long-term tracking process. Besides, we also achieved the highest mIoU (Intersection over Union) score of 92.4%, indicating accurate instrument localization, which significantly surpassed PAWSS (53.4%, $P < 0.0001$) and DeepSORT (84.3%, $P = 0.0093$). This superiority is attributed to our consideration of both orientation and insertion direction, which neither PAWSS nor DeepSORT considers. The typical tracking screenshots of our model are shown in Fig. 3D. We also conducted experiments on several 1-minute video clips under typical dynamic scenarios (varied FoV/instrument motion velocity, smoke/blood contamination, etc.). Our method achieved at least 94.9% accuracy with 86.8% mIoU, whereas DeepSORT dropped to 66.8% tracking accuracy and 75.7% mIoU (see Table S2 and Fig. S3). The stereo instrument depth estimation achieved an accuracy of $0.8 \pm 0.5 \text{ mm}$ over a 120 mm workspace. Our tracking framework achieved real-time performance at an excellent 76 fps while simultaneously tracking multiple instruments, with negligible time cost for additional instruments. The above results demonstrate that our tracking framework can achieve a robust and accurate tracking performance, hence providing a solid foundation to further STRUM-based operations in long-horizon

surgical tasks.

Surgical phase recognition. We have observed that during the surgery, surgeons have different preferences for camera views and uterine manipulations at different phases of the operation. For instance, in the early preparation stage, surgeons need a wider FoV and a faster laparoscopic camera or uterine movements to assess the overall surgical procedure. In the middle stages of hysterectomy, precise operation within a stable view range is essential, and surgeons prefer a small-scale FoV and slow movements. To enable our robots with intelligent surgical phase understanding capability, we have established a surgical phase recognition algorithm and built it into the software modules of our OSFA system. The integration of multiple intelligent schemes can jointly identify surgical phases in real-time and assist in the optimization of control algorithms, providing optimal robot motions with proper maneuver speed to achieve safer cooperative controls in robotic hysterectomy.

Generally, the total laparoscopic hysterectomy (TLH) procedure comprises seven phases [32], while we focused on the four most critical stages in our experiments: *Preparation*, *Dividing Ligament and Peritoneum*, *Dividing Uterine Vessels and Ligament*, and *Transecting the Vagina*. As shown in Fig. 3F, we collected and annotated 21 full-length surgical videos, with a total video duration of 0.76, 7.56, 5.11, and 2.99 hours (i.e., the median durations are 59.18, 1160.81, 879.08, and 419.92 seconds) for each phase. We employed the Trans-SVNet algorithm [33] for phase recognition, which is designed based on the Transformer architecture and

combines spatial and temporal information, achieving accurate predictions of the surgical phases. Fig. 3G showcases typical examples of successful predictions from the offline training on our collected dataset. Additionally, the quantitative results of phase recognition, as shown in the confusion matrix in Fig. 3H, present precise recognition results with overall accuracy exceeding 85%, revealing its promising potential in guiding control scheme selections in clinical trials. We further formalize the operation preferences into phase-specific parameters, specifying the optimal field-of-view regions (50%, 50%, 40%, and 60% of the image width and height for Phases 1–4) and corresponding laparoscope movement speeds ($\times 1.0$, $\times 1.6$, $\times 2.0$, and $\times 1.2$ for Phases 1–4), which are integrated into our system to guide camera control throughout different surgical phases. Although a small portion of misclassifications may lead to suboptimal field-of-view settings or motion speeds, our system is designed with phase-specific parameter ranges and smooth transition mechanisms, ensuring that such deviations do not adversely affect the surgical workflow or compromise operational safety. The integration and validation results on cadaver experiments are presented in Movie S2.

Experimental demonstrations of LoCC: Phantom study

To enable long-horizon solo surgeries with autonomous robotic assistance using our intelligent modules, we hereby introduce a framework consisting of five levels of cooperative control (LoCC) for minimally invasive surgery, providing a structured pathway toward higher levels of clinically viable autonomy. Each level represents a progressive enhancement in either the system's autonomous capabilities or its direct interaction with the primary surgeon through human-in-the-loop control, while adhering to the design principle of minimizing external hardware to ensure seamless clinical integration without explicitly increasing the surgeon's burden. In this work, we instantiate the five LoCC levels in the context of laparoscopic hysterectomy as a representative surgical task. Their characteristics are depicted in Fig. 4A.

We integrated the five LoCC levels into our OSFA system and evaluated their functionality and efficiency through experiments on a phantom platform, and the experimental setup is shown in Fig. 4B. In the experiments of all five levels, we mimicked the laparoscopic hysterectomy task in an abdominal cavity phantom. As evidenced in Fig. 4C, the typical procedures begin with an initial global view of the whole operating scenario. Then, in Uterus Task-1, the uterus is moved right and up by the manipulation rod to provide space for instrument operation, followed by Laparoscope Task-1, in which the laparoscope tracks the primary instrument to the left side of the uterus and zooms in to maintain a close view of the operative area. After completing the left-side procedure, during Uterus Task-2, the uterus is moved left, and in Laparoscope Task-2, the laparoscope follows this movement to the right side and zooms in again. Finally, in Laparoscope Task-3, the laparoscope zooms out to restore the global view, completing the whole procedure. The errors of the laparoscope trajectories and the manipulated uterus of each task are shown in Fig. 4D–E. The errors are calculated based on the positions within the image plane relative to the predefined target. The corresponding time durations are presented in Fig. 4F–G, representing the elapsed time from the initial moment until the target position is reached.

Level 0. Assistant manual control. In the traditional operating room for laparoscopic hysterectomy procedures, two assistant

surgeons are needed to assist the primary surgeon. One holds the laparoscope to adjust the FoV, while the other operates the manipulation rod to position the uterus, assisting the primary surgeon's operation around the uterus. Both assistants must stay focused on the ongoing surgical scenario and the primary surgeon's operation from the view displayed on the monitor. Experienced assistants generally cooperate more smoothly with the primary surgeon, whereas junior assistants often require on-site instruction for specific maneuvers, potentially compromising surgical efficiency.

To validate the performance of LoCC-0 under the phantom-based scenario, three task-experienced staff with surgical knowledge and expertise in surgical robotics were invited to conduct the simulated experiments. As shown in Fig. 4F–G, the manual operations of the experienced assistant were fast and direct. In contrast, the robotic system under LoCC 1–4 operated with constrained movement speed due to safety considerations. As a result, LoCC-0 achieved the fastest completion times in all laparoscope control tasks with 41, 26, and 16 seconds, respectively. The assistant also demonstrated relatively fast performance in the uterus manipulation tasks, with durations of 11 and 4 seconds. However, there still exist potential gaps in the multi-person cooperative task, in which the position of the manipulated uterus exhibits maximum errors of 114.1 and 178.9 pixels in the x and y directions, as reflected in Fig. 4E. Additionally, manual operation possesses the nature of sharp trembling during the intermediate motions, which may impair the surgical safety. The complete procedure of LoCC-0 is demonstrated in Movie S3.

Level 1. Assistant tele-operation control. At this level, robots are introduced to perform tasks traditionally handled by assistant surgeons, relieving humans of tedious operations and enhancing the stability of movements. In the proposed OSFA system, a laparoscope-holding robot arm and a uterine manipulation robot have been developed to replace the two assistant surgeons. However, assistant surgeons are still needed to tele-operate the robots based on their understanding of the surgical scene or the primary surgeon's commands.

As shown in Fig. 4D and Movie S4, LoCC-1 exhibits large and frequent deviations in the error curves (i.e., x -axis error of 110.2 ± 104.3 pixels) during instrument movement, reflecting the assistant's misinterpretation of the surgeon's instructions. This leads to repeated adjustments and significantly prolonged task execution times of 133, 71, and 36 seconds across the three laparoscope tasks, representing increases of 224.4%, 173.1%, and 125.0% compared to LoCC-0. While this delay is partly due to the inherently slower speed of robot movements, which are deliberately constrained for safety reasons, robotic assistance still offers clear benefits. The robots' consistent and tireless performance significantly reduces the assistants' workload and enhances the overall stability of the procedure. To address both the robot speed limitations and the communication gap between surgeon and assistant, while also aiming to further minimize human resource demands, we explored more advanced and efficient control strategies in the subsequent LoCC levels.

Level 2. Simple commands-guided control. In the operating room, human resources are exceptionally valuable and should be utilized efficiently. To reduce reliance on assistant surgeons for routine tasks, LoCC-2 introduces a predefined command-based interaction enabling semi-automated control. Each directional movement of the uterus manipulator and laparoscope zoom

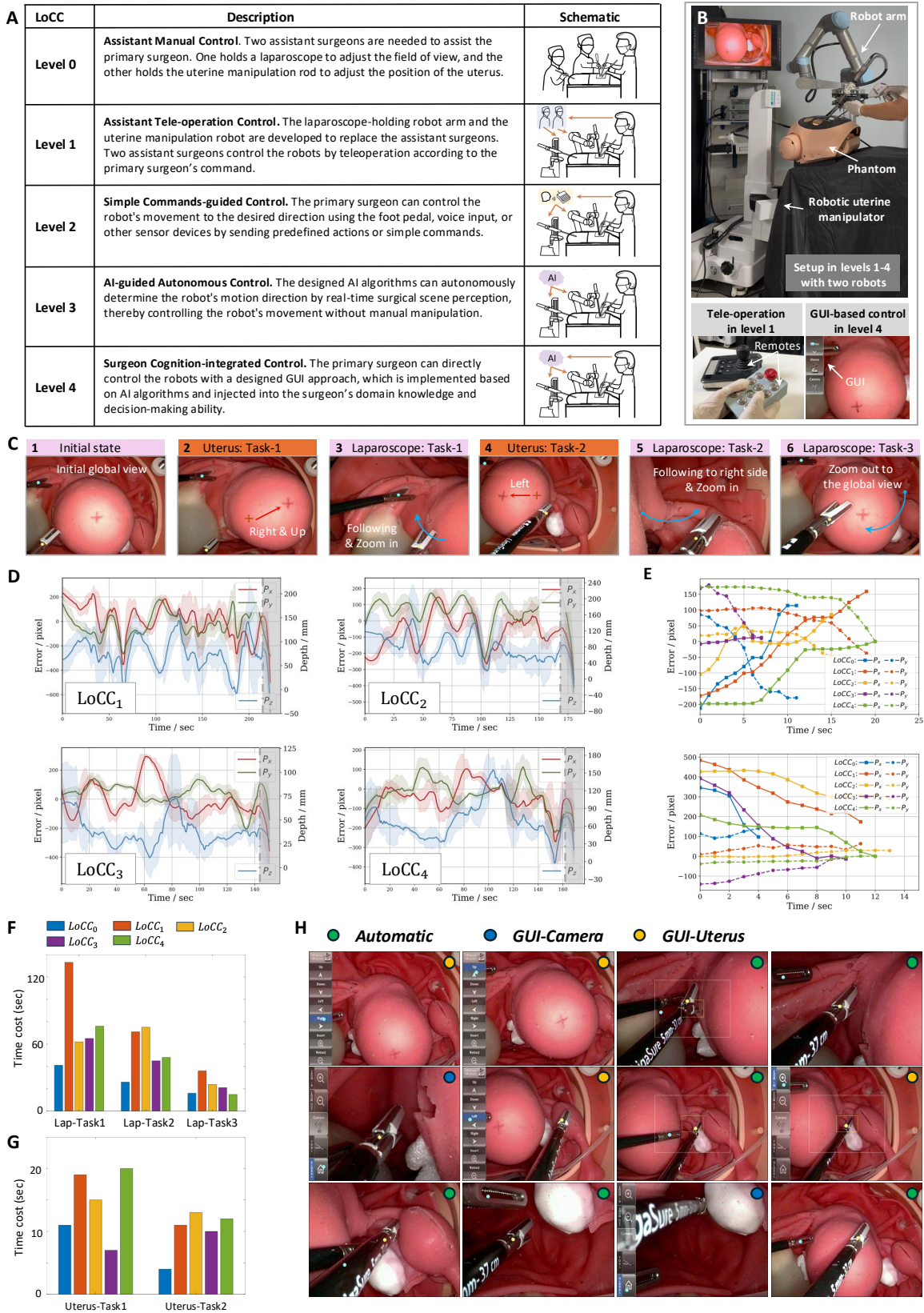


Figure 4 Definition of the five Levels of Cooperative Control (LoCC) and corresponding phantom experiments using our OSFA system. (A) Overview and schematics of the five LoCC levels. (B) Experimental setup of the OSFA system on the phantom platform. (C) Simulated laparoscope and uterus manipulation tasks within the abdominal cavity. (D) The x/y pixel errors and depth of the primary instrument during laparoscope tasks across LoCC 1-4. (E) The x/y pixel errors in the image plane between the controlled and target uterus positions in uterus Task 1 (top) and Task 2 (bottom). (F) Comparison of task completion times for laparoscope control across five LoCC levels over three task types. (G) Comparison of task completion times for uterus manipulation across five LoCC levels over two task types. (H) Comparison of task completion times for uterus manipulation across five LoCC levels over two task types. (I) Screenshots of the simulated tasks, with the top-right dot indicating the current system status.

level corresponds to a distinct voice command, activated by pressing a foot pedal and deactivated upon release, allowing hands-free operation in surgery.

Fig. 4D shows that LoCC-2 enables the laparoscope to achieve smoother and more direct trajectories toward the target points, exhibiting lower error levels compared to LoCC-1. Specifically, the mean x -axis error for LoCC-2 is 97.5 ± 69.5 pixels, substantially lower than the 110.2 ± 104.3 pixels under LoCC-1. Similar improvements are noted in the y -axis and depth, indicating that LoCC-2 aligns the laparoscope motion more efficiently with the surgeon's intent. As shown in Fig. 4F and 4G, the execution time is also reduced in most tasks. Specifically, the three laparoscope tasks took 62, 75, and 24 seconds, respectively, with an average of 53.67 seconds and 43.3% shorter than LoCC-1 (94.67 seconds). For the two uterus manipulation tasks, the execution times were 15 and 13 seconds, reflecting a 6.7% reduction in average compared to LoCC-1. However, it should be noted that operating rooms are densely equipped with medical devices, including electrically driven tools operated via foot pedals. Hence, introducing additional devices may disrupt surgical workflow and introduce potential safety risks. The video of this experiment is Movie S5.

Level 3. AI-guided autonomous control. With advances in learning-based visual analysis, surgical scene perception has become essential for developing intelligent operating rooms. At LoCC-3, our goal is to leverage AI-based visual parsing to interpret online surgical scenes and actions, enabling robots to autonomously execute specific tasks through AI-generated commands. In the OSFA system, a detector precisely localizes surgical instruments, as described in the instrument detection section. Based on such information, the laparoscope autonomously tracks the main instruments to keep them centered in the FoV, ensuring optimal visualization. Meanwhile, the uterus is autonomously repositioned away from the main instrument to provide sufficient operating space, facilitating easier surgical access and direct cutting manipulation around the cervix.

As shown in Fig. 4D-G, the intuitive and scene-adaptive automatic control strategy enabled LoCC-3 to achieve a comparable mean x -axis pixel error of 92.4 ± 70.3 pixels to LoCC-2, while significantly reducing execution time across most tasks compared to LoCC-1 and LoCC-2. Specifically, the execution times for the three laparoscope tasks were 65, 45, and 21 seconds, with task 2 exhibiting reductions of 36.7% and 40% compared to LoCC-1 and LoCC-2, respectively. In the uterus manipulation task 1, the time dropped by 63.2% and 53.3%. The video of this experiment is Movie S6.

Level 4: Surgeon cognition-integrated cooperative control. In LoCC-3, the developed autonomous control scheme is restricted to certain ranges of surgical tasks and may yield unintended behaviors due to compounding prediction errors, potentially leading to catastrophic outcomes in cases of AI failure during online operations. Therefore, a more advanced and clinically feasible LoCC-4 is hereby introduced. At this level, AI continuously generates low-level control commands for the robots, while the primary surgeon supervises key task nodes and retains ultimate authority, correcting robot motions through an intuitive interactive GUI informed by surgical expertise and cognition. This new level of cooperative control paradigm can enhance both the manipulation accuracy and safety, especially applicable to long-horizon and complex surgical tasks automation.

Specifically, we retain the AI-based workflow from LoCC-3 while introducing a user-friendly GUI that allows the primary surgeon to interact directly with the system during surgery, as detailed in the aforementioned section concerning the GUI design. The surgeon's domain knowledge and high-level decision-making capabilities can be coherently conveyed to the robots, enabling seamless control without the need for any additional equipment. This LoCC-4 acts as the glue of multiple technical modules, which combines AI-empowered semi-autonomous control with online human-robot interaction, consequently enhancing task convenience, manipulation efficiency, and surgical reliability in clinical rooms, while alleviating the operational burdens of surgeons.

As shown in Fig. 4D-H, LoCC-4 achieves the best overall performance among the control strategies, balancing task efficiency, motion stability, and practical applicability. The execution times for the three laparoscope control tasks were 76, 48, and 15 seconds, while those for the two uterus manipulation tasks were 20 and 12 seconds. Notably, these durations include the time required for the surgeon to operate the instrument to interact with the GUI, as evidenced by the relatively flat error curves during the first 5 seconds. Although this control mode results in slightly longer execution times, the human-in-the-loop approach ensures greater precision and operational stability. As shown in Fig. 4D-E, the laparoscope accurately tracked the main instrument with mean x/y -axis error of 70.7/60.3 pixels, achieving the lowest overall error across all LoCC modes. For uterus manipulation, the surgeon could directly control uterine movement through the GUI, achieving near-zero error in both x and y directions upon task completion. LoCC-0 achieves the shortest task execution time but requires three experienced co-ordinated operators, making it resource-intensive. In contrast, among the robot-assisted strategies (LoCC-1 to LoCC-4), LoCC-4 achieves the most efficient task execution with only a single surgeon using an intuitive GUI-based interface, providing a more efficient, precise, and scalable solution for clinical application. These outcomes highlight the advantages of our proposed AI cooperative control strategy in enhancing surgical performance and overcoming challenges from traditional control paradigms. The video of this experiment is Movie S7.

Cadaver demonstration of robotic laparoscopic hysterectomy

We conducted cadaver experiments to test the feasibility of our OSFA system for robotic hysterectomy at Jockey Club Minimally Invasive Surgical Center of The Chinese University of Hong Kong, Prince of Wales Hospital, HKSAR, China. A surgeon with over thirty years of experience performed the entire procedure on a female cadaver, which is shown in Fig. 2A. After initial manually positioning the robotic laparoscope and uterus manipulator, we activated the OSFA system and switched the laparoscopic state into the *AUTOMATIC* state, enabling a fully automatic laparoscope control to actively track the main instrument and provide the appropriate FoV for the surgeon's operation. Since the main operative instrument varied across surgical phases, e.g., LigaSure in phase 2/3 and electric hook in phase 4, our system endows convenient selection of the surgeon's main instrument category, thereby benefiting a precise tool tracking clinically. Unlike the frequent adjustments of the laparoscope, the uterus manipulator was manipulated intermittently as needed.

The entire cadaver-based laparoscopic hysterectomy utilizing our intelligent OSFA system lasted approximately 79 minutes,

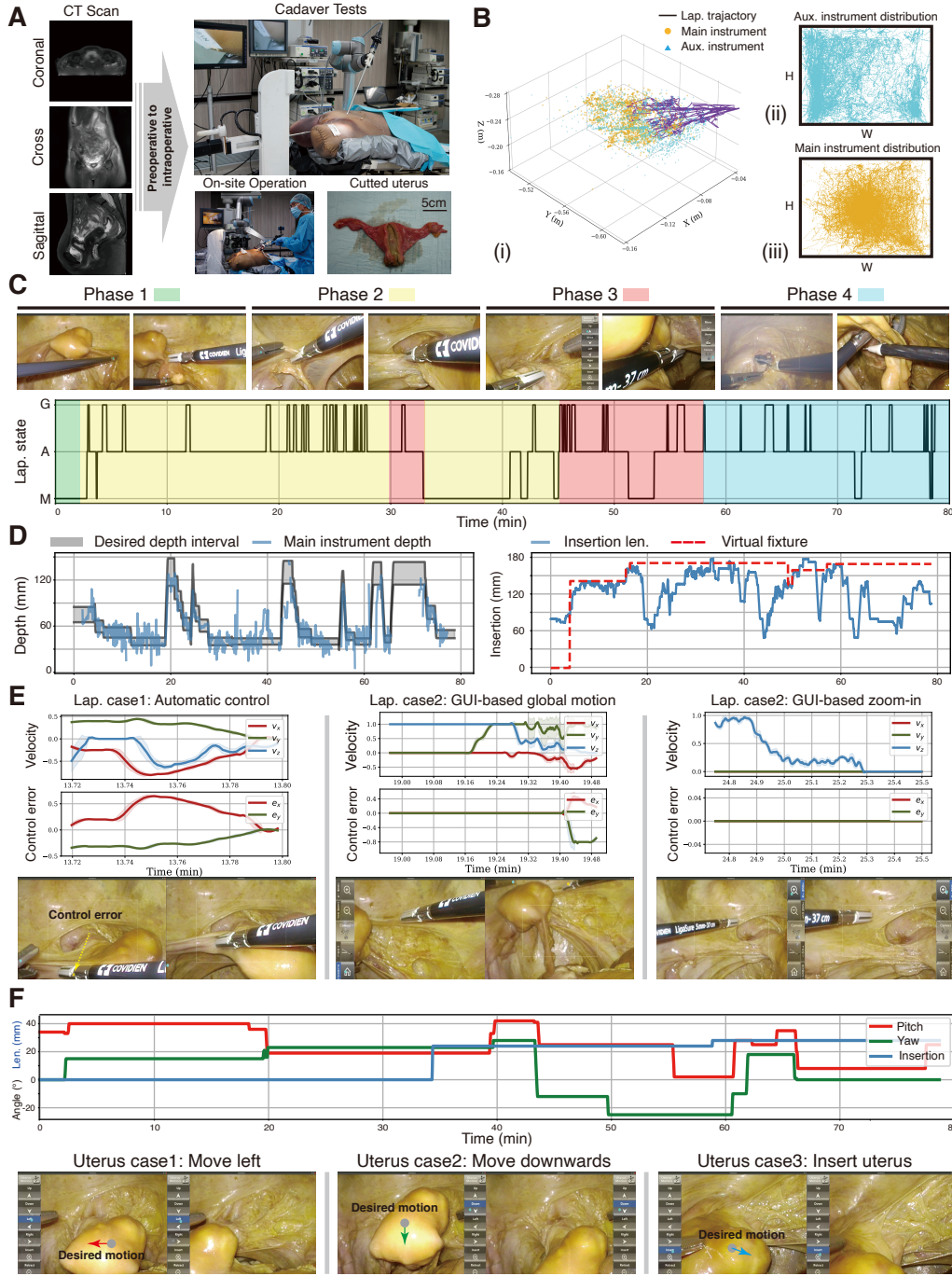


Figure 5 The results of the OSFA system in performing cadaver hysterectomy. (A) Pre-operative CT data and intra-operative OSFA system for a successful solo-hysterectomy. (B) The instruments trajectory and the laparoscope trajectory, along with the image re-projection of the auxiliary instrument and the main instrument. (C) Laparoscope states throughout the cadaver study, as well as corresponding snapshots of the surgical procedure. (D) Performance of the depth controller in maintaining the main instrument depth within a proper depth interval (left) and the ZVF constraint (right). (E) The typically selected laparoscope control clips from the cadaver study with their corresponding enlarged control velocity and control errors. (F) The top shows the pitch, yaw, and roll joint information of uterus manipulator during the cadaver study. The bottom part of the three rows shows the separate DOF control in the left, down, and insertion directions.

as shown in Movie S8. Note that the cadaver had a uterine myoma, increasing the complexity of the surgery as shown in Fig. 5A. The 3D instrument trajectory and corresponding projections throughout the procedure are presented in Fig. 5B. It shows the 3D spatial positions of the laparoscope and instruments, and the 2D pixel locations of the auxiliary and main instruments in the surgery, respectively. Though the operations are conducted in different regions, the main instruments are consistently located in the central area of the image, which demonstrates the

effectiveness of our automatic FoV control strategy. In contrast, auxiliary instruments exhibit higher spatial distributions along the lateral edges of the GUI panel, indicating their frequent usage in HMI. The inter-laparoscope state and corresponding screenshots for each phase are shown in Fig. 5C. Besides, Table. 1 summarizes the system's overall performance during the solo-hysterectomy. For laparoscope control, the system operated in automatic mode (A) for the majority of the time, with 70.4% in automatic mode and 10.9% in HMI mode (G). Manual mode

was employed in only 18.7% of cases, including preoperative setup (2.5%), situation where laparoscope was positioned too close to tissue (6.8%), and procedures requiring highly dynamic view adjustments (9.4%), demonstrating the feasibility of the proposed semi-automatic strategy and greatly improving the FoV adjustment efficiency. Besides, the GUI interaction accuracy, defined as the proportion of successful interactions relative to the total attempts, was 93.6%. The observed failures were primarily due to incorrect instrument tracking, which disrupted the interaction. These results validate the overall effectiveness of our tracking-based approach.

Fig. 5D presents the depth of the main instrument and its desired interval. The instrument was consistently maintained within the relatively desired depth interval despite its frequent depth changes as the surgeon adjusts it via GUI interaction as needed. Besides, the laparoscope's insertion length relative to the trocar point always remains below the set threshold throughout system operation in semi-automatic mode, thus ensuring surgical safety by avoiding the laparoscope being too close to the tissue. Fig. 5E illustrates three typical semi-automatic laparoscope movements. Fig. 5E1 illustrates the automatic FoV control process, when the main instrument leaves the central area, the laparoscope can shift the FoV smoothly within 5 seconds to bring the projection of the main instrument on the image back inside the target yellow rectangular area, and the corresponding laparoscope movement slows down as the target position approaches. Fig. 5E2 shows a successful automatic return process from the narrow local view to the large global view via GUI interaction, followed by a 2D automatic view adjustment for tracking main instruments with $v_z = 0$. Fig. 5E3 shows a successful *zoom-in* action, where a deeper magnified view is achieved by successive clicks of the *zoom-in* button for finer operation, with zero velocity in the xy directions. The control velocity and the control errors for laparoscope manipulation throughout the entire cadaver study are shown in Fig. S4.

In terms of uterus control, the robot predominantly lifted the uterus autonomously with 92.1% automation (H), while GUI interactions facilitated necessary adjustments, contributing to the successful solo hysterectomy without extra assistants. Fig. 5F illustrates the uterus motions in 3-DOF throughout the cadaver study. The control of the robotic uterine manipulator was not as frequent as the laparoscopic control, with the GUI buttons activated 27 times: 14 for the left/right movement, 9 for the lifting/pulling down, and 4 for the insertion/retraction. Typical screenshots of uterus control demonstrate the surgeon moving the uterus directly through HMI to facilitate further surgical operations, such as shifting left, pulling down, etc. Movie S9 shows a detailed recording of each phase throughout the entire cadaver study.

Discussion

Achieving a higher level of both surgical autonomy and efficiency requires sophisticated and intelligent cooperation between the primary surgeon, AI, and robots. By improving surgical precision, stability, and dexterity, robotic systems expand the scope of what one single surgeon can achieve in modern operating rooms, paving the way for a more intelligent and cooperative autonomous surgical paradigm. In this work, we introduce our one-surgeon-four-arm robotic system that is capable of collaboratively and autonomously realizing a long-horizon hysterectomy in clinical settings. Based on the development of our AI-empowered cooperative paradigm, the system, for the first

time, enables one primary surgeon to complete the entire TLH without any human labor, thereby improving the operational efficiency and consistency by eliminating discrepancies from the linguistic and empirical coordination of surgical assistants. The success of our solution hinges on two core components. The first one is our hierarchical architecture that decouples the high-level decision-making and low-level automated control. At the high-level stage, task decision is shared between the AI system and the surgeon, allowing the surgeon to intervene at any time to transcend the AI-based control. The second one is our cooperative paradigm that can harmoniously glue multiple discrete algorithmic modules. With our precise instrument tracking approach and intuitive instrument mouse-based GUI interaction, the paradigm can be smoothly integrated into our devised robotic system without any pre-deployment of additional sensors. Even under highly dynamic surgical environments, our cooperative manner can enable the surgeon with extra surgical maneuverability in a cost-effective way, paving a realistic road map towards clinically feasible surgical intelligence for long-horizon task automation.

At the heart of our system is the conceptualization of the roles of surgeon-AI-robots and the introduction of a multi-level cooperative framework (LoCC) for multi-arm operations, which is designed to support long-horizon surgical procedures. In laparoscope control, our transition from task-level to procedure-level autonomy is informed by enhancing the efficiency and completeness of entire surgical procedures, including scene understanding and seamless surgeon-AI-robots interaction. Prior works have primarily focused on simplified tasks [34, 35, 36, 37, 38, 39], with evaluations conducted exclusively on phantom setups evaluation [17, 40] or within low-fidelity and toy-scale environments [41, 36]. These efforts fall within the LoCC-2 paradigm and do not fully exploit surgeon-AI-robots cooperation across long-horizon clinical surgery. Such approaches face inherent limitations in clinical practice, where surgical scenes are dynamic, explicit tracking sensors are unavailable, and context-aware zoom control remains a critical unsolved problem. Our work advances beyond these constraints by introducing cooperative assistance from LoCC-1 to LoCC-4, validated in a pre-cadaver phantom case study. Unlike prior methods, the OSFA system retains the surgeon as the primary operator while providing adaptive, context-aware support throughout the entire workflow. This balanced integration of autonomy and surgeon control enables robust handling of complex scenarios and effective adaptation to the variability and unpredictability of real surgical environments.

For the cadaver study on a female subject, we adopted the LoCC-4 paradigm to conduct the first entire solo-hysterectomy worldwide. The procedure was successfully completed on a uterus weighing 86g, substantially larger than the typical postmenopausal range (50~60g), representing a particularly challenging scenario for conventional practice and highlighting the robustness of the system. This framework enables the surgeon to delegate assistant tasks to the AI controller while remaining ready to intervene when necessary. Conceptually, this reduces the surgeon's cognitive load by integrating two automated assistants that provide consistent laparoscope control and uterine countertraction, allowing the surgeon to concentrate on the primary manual tasks. The system can be set up within 5 minutes without pre-deployment of additional sensors and imposes a relatively low learning burden compared with conventional robotic systems. In our study, the surgeon became proficient with the

Table 1 Cadaver hysterectomy. The surgery was carried out on the female cadaver using our OSFA system with Level 4 cooperation. The left/right number in the GUI-referenced column represents the corresponding number of activated human-in-the-loop controls for laparoscope/uterus manipulator.

Surgical Phases*	Num. of GUI activation	Acc. of GUI activation (%)	GUI time (min)	Completion time (min)	Ratio of lap. status M/A/G †(%)	Ratio of uterus status H/G †(%)
Phase 1	0/2	100	0/0.4	2.0	100/0/0	81.0/19.0
Phase 2	21/12	94.3	3.8/2.8	39.6	22.8/67.6/9.6	92.9/7.1
Phase 3	11/2	92.9	1.71/0.3	16.4	14.8/74.8/10.4	98.1/1.9
Phase 4	14/11	92.6	3.1/2.7	20.7	2.9/82.2/14.9	87.0/13.0
Total	46/27	93.6	8.6/6.2	78.7	18.7/70.4/10.9	92.1/7.9

* Phase 1: Procedural preparation; Phase 2: Ligament division; Phase 3: Vessel division; Phase 4: Vaginal transection.

† Lap. status: M - Manual mode, A - Automatic mode, G - GUI mode; Uterus status: H - Hold mode, G - GUI mode.

system after two 30-minute phantom trials before proceeding to the cadaver case, contrasting with the reported average of approximately 20 hours required to reach comparable proficiency with the da Vinci system [42]. Besides, the durations of the individual phases in TLA with da Vinci system were 19.1 ± 15.7 , 78.6 ± 9.4 , 83.7 ± 17.5 , and 4.2 ± 1.5 minutes [43]. In comparison, the corresponding durations with our OSFA system were 2.0 minutes (9.6x faster), 39.6 minutes (2.0x faster), 16.4 minutes (5.1x faster), and 20.7 minutes (0.2x, i.e., longer), respectively. Our results demonstrate that the evolved LoCC-4 mode enhances the responsiveness and adaptability of both laparoscope and uterine manipulator control, thereby improving the overall efficiency and consistency of solo hysterectomy procedures.

Our OSFA system may yield suboptimal performance concerning its autonomous control when the laparoscope is excessively zoomed in and the handheld main instrument moves out from the FoV, in which the narrow working space between the laparoscope and surrounding tissue makes it difficult to re-track the instrument autonomously. In addition, considering that our system intends to provide stable AI-based control of robots, when surgeons frequently adjust zoom levels or perform operations involving large and rapid instrument motions, the current automatic system may fall short of accommodating personalized and dynamically evolving view preferences. In such cases, manual mode should be triggered, and the primary surgeon needs to take control of the entire system. For overall stability, the system now performs a passive switch to manual mode, enabling instrument adjustment and repositioning. Subsequently, the automatic control mode can be reactivated. To further enhance the system's autonomy and reduce surgeon's cognitive burdens, it is worth improving the self-assessment capability of the system, which can intelligently indicate the suboptimal conditions and actively switch to the manual control model for surgeons. Besides, language-based interaction between surgeon-robot possesses the potential of enhancing the interaction efficiency, and its integration into our system should be further investigated.

The concept of AI-empowered LoCC has the potential to transform not only laparoscopic hysterectomy but also a wide range of medical robotic applications where multiple robotic arms provide cooperative assistance. Similar assistive tasks are technically feasible and could be advanced to in vivo trials in the

near future. Such cooperative robotic systems can anticipate the surgeon's needs and deliver real-time support. As these surgical copilots continue to evolve, the establishment of comprehensive ethical guidelines and regulatory frameworks—through joint efforts among clinicians, researchers, industry stakeholders, and regulatory authorities—will be essential. The roadmap will push the boundaries of surgeon-AI-robots collaboration while ensuring safety and efficiency, and ultimately improving patient care.

Apart from the OSFA hardware system, we proposed a novel GUI-based interaction mode for vision-based collaborative control of these two robots, providing a notable advantage by eliminating the pre-deployment of additional sensors. This adaptability is of particular significance in existing hysterectomy setups, as it minimizes the requirement for procedural modifications or hardware upgrades, thereby facilitating seamless integration of intelligent surgeon-AI-robots cooperative control. To realize the real-time interaction with GUI buttons, we proposed a maskless multi-instrument tracking strategy, which serves as the foundation for the Instrument-Mouse interaction. The main instrument's tracking enables FoV adjustment, while other tracked auxiliary instruments allow direct surgeon-robots interaction through the GUI, alleviating the need for communication with assistants.

Methods

Robotic platform

The laparoscope motion is constrained by the Remote Center of Motion (RCM) mechanism, providing it with 4 DOFs, as shown in Fig. 2E. These include pan and tilt motions (1st and 2nd DOFs) for pivoting around the trocar insertion point, resulting in transitional movement along the image. The 3rd DOF allows insertion and retraction along the longitudinal axis to resize the FoV, and the 4th DOF enables spinning rotation around the insertion axis, leading to image center rotation. The robotic laparoscope manipulator consists of a 6-DOF UR5 robot driven by a soft-RCM mechanism and a stereo laparoscope system.

In addition, there are three main DOFs for uterine motion control: the yaw and pitch DOFs are used to move the uterus in the 2D image plane; the third insertion/retraction DOF along the uterine normal axis is used to control the stretching of the uterus, as shown in Fig. 2F. Specifically, the yaw motion is achieved by

the transmission of two gears and the gear rim. The pitch motion is realized by the single-motor bilinear-guided (SMBG) mechanism, where the manipulation rod passes through two sliders that move vertically along linear tracks. The displacement of these sliders is controlled by two pulleys with different numbers of teeth, enabling a stable mechanical RCM mechanism that supports precise and safe uterine manipulation. Besides, the uterine manipulator is equipped with a handle that integrates a force sensor for motions in 3-DOF. Through admittance control, it allows for intuitive and efficient placement of the robot's RCM point at the cervix, enabling rapid preoperative positioning [27].

Human-machine interface

The OSFA software system contains the following four main subtasks: Instrument Mouse Perception (IMP), Overlay Graphical Interface (OGI), Laparoscopic Manipulator Control (LMC), and Uterine Manipulator Control (UMC). The IMP component perceives the surgical instruments in real time. The STRUM interprets the instruments staying in a specified image area for a period of time as a mouse click, enabling the surgeon to interact directly with the OGI component in a human-in-the-loop manner by moving the instruments to control the robotic laparoscope/uterine manipulator as needed. Then the LMC and UMC components convert the motion targets into control speeds for the corresponding robot joints.

GUI State Design for Laparoscope Control: Based on the surgeon's feedback and clinical analysis, we designed the robotic laparoscope control system with the following four states: 1) *AUTOMATIC*: guiding the laparoscope to keep the instrument tip in the center region of the view while maintaining the proper depth interval. This state corresponds to the automatic 3D spatial control of the laparoscope, which contains the pan, tilt, insertion, and retraction laparoscopic camera functions; 2) *ZOOM*: guiding the laparoscope to provide the desired target magnification. This state corresponds to the human intervention of the insertion and retraction camera functions. 3) *GLOBAL*: guiding the laparoscope automatically to provide the desired global view. This state means that the laparoscopic camera will return to the initial or preset global position in the surgical scene. 4) *GLOBAL-HOLD*: This state is similar to *AUTOMATIC* state, but without depth control feature. The surgeon can intuitively treat auxiliary instruments as the computer mouse, namely STRUM, to activate the laparoscope panel selection interface (putting the instrument in a specific image region corner) and modify the laparoscope state accordingly by virtually clicking the corresponding button (staying in the button area for a certain duration), as shown in Fig. S2.

GUI State Design for Uterus Control: The control of the robotic uterine manipulator has a similar mechanism to the laparoscopic control and the difference is that separate buttons are designed to control the uterine moving direction, i.e., up and down, insertion and retraction, etc., as described in Table S1. The system contains *HOLD* and *KINETIC* states. It is mostly in the *HOLD* state to maintain the current uterine pose and only switches to the *KINETIC* state when the action button is clicked, where the uterine moves at a constant speed in the corresponding direction, with a gradient color on the button indicating the extent of movement. It returns to the *HOLD* state when the instrument is out of the button area.

Instrument motion tracking

To facilitate precise HMI using instruments as a virtual mouse, we develop a robust and highly accurate instrument motion tracking framework tailored for the dynamic surgical scenes. Our approach integrates a multi-task orientated instrument network (MTOI-Net) that jointly predicts the rotated box and its upper-left index for fine-grained localization. To ensure robust tracking performance over time, we introduce a long-term memory-based tracker framework (LTMT) that maintains temporal continuity and recovers from missed or false detections. This design significantly improves tracking reliability and precision under challenging intraoperative conditions.

Network structure: A global schematic of MTOI-Net is shown in Fig. 6. The encoder comprises CSPDarknet as the backbone, convolutions, batch normalization, and spatial pyramid pooling. The decoder upsamples the features and concatenates the shadow backbone features of 1/8, 1/16, 1/32, which are then fed into CSP bottleneck layers and a 5x5 convolution layer with batch normalization. To uniquely determine the instrument position, separate heads in multiple scales are employed for predicting the category c , rotated box $p = [c_x, c_y, w, h, \theta]$, and corresponding upper-left ul index (i.e., shaft direction), where c_x, c_y is the center position, w, h is the box size, $\theta \in [0, 2\pi)$ is the rotation angle following the long-edge representation. We use the circle smooth label (CSL) with 1° interval to encoding θ in a continuous and differentiable representation [44] and formulate it into a classification problem. The final dense layer of category, upper-left, and θ head uses the softmax activation function, while the rotated box head utilizes the sigmoid function for regression. As depicted in Fig. 6A, the $pt1$ is calculated with the center point and the $vector1,2$ associated with θ , while other points are obtained uniquely in a clockwise manner, therefore the sign of θ results in the different shaft directions with the same pt number, hence we establish a decoding strategy for the discontinuity caused by the sign of θ between the ul label and insertion direction. The whole model is trained with a combined loss $\mathcal{L}$ under multiple scales:

$$\mathcal{L} = \sum_{y \in \{c, \theta, ul\}} \sum_k \mathcal{L}_{CE}(y^k, \hat{y}^k) + \mathcal{L}_{JA}(b^k, \hat{b}^k) \quad (1)$$

where y^k and $\hat{y}^k$ represent the ground-truth (GT) label and the network prediction of task $\{c, \theta, ul\}$ at stage k respectively. $\mathcal{L}_{CE}$ is the cross-entropy loss. $\mathcal{L}_{JA}$ is the Jaccard loss between the GT box b^k and corresponding prediction $\hat{b}^k$ at stage k .

Long-term instrument tracking: The instrument tracking logic is illustrated in Fig. 6B. It enables robust multiple instrument tracking for stable STRUM-based interactions, even with occasional wrong or missing detections from the raw MTOI-Net predictions. It comprises two separate trackers: one for the continuous orientated box p and another for the categorical shaft direction. For orientated box tracking, considering the instrument mainly operates on a relatively fixed area but with rapid velocity under dynamic surgical scenarios, we propose a memory-based matching metric named Intersection over Target (IoT) that considers the historical locations, which compensates for cases where the

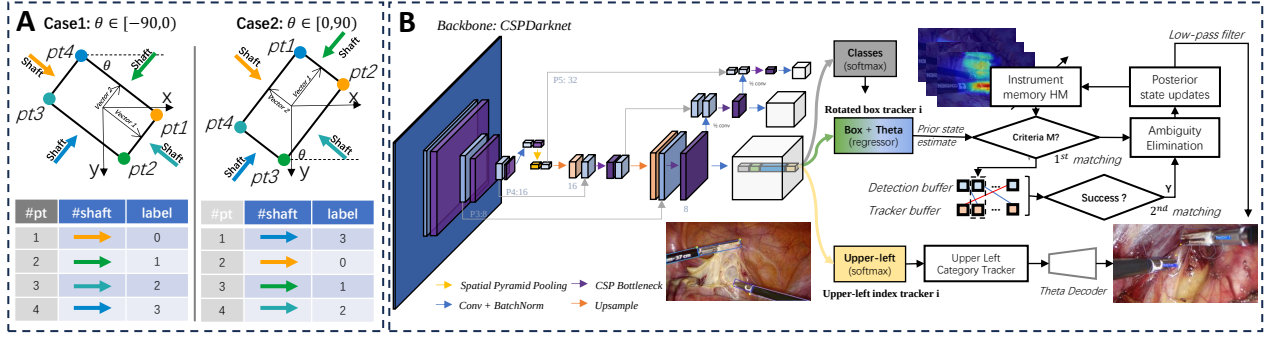


Figure 6 Instrument motion tracking model. (A) Discontinuity illustration of upper-left index caused by the sign of θ under the long-edge box representation, and Decoding rules of the label correspond to the sign of the θ and the point number (#pt). The shaft and corresponding upper-left point are indicated by the same color. (B) The overall framework of LTMT consists of an MTOI-Net and a two-stage memory-based tracking pipeline for the rotated box and the corresponding upper-left label, which can uniquely locate the instrument's position and insertion direction in real-time.

IoU metric in adjacent frames is insufficient for tracking:

$$\text{IoT}(\text{box}_0) = \frac{\sum_{i,j \in \text{box}_0} \mathbb{1}(\text{HM}_{ij} > \sigma_{hm})}{w_0^{\text{box}} \cdot h_0^{\text{box}}}, \quad (2)$$

$$\text{HM} = \sum_{t=-T}^{-1} \mathbb{1}(\text{box}_t) \cdot \gamma^t.$$

where HM is a time-weighted heat map of instrument locations in the past T timestamps, weighted by a forgetting factor γ . $\mathbb{1}$ is the indicator function. We build a two-stage tracking process by introducing a conditionally updated Kalman filter with states $\mathbf{x} = [p, \dot{p}]^T \in \mathbb{R}^{10}$. The main judgment for the first-round matching process is to match the Kalman filter predicted state and compare the IoT and IoU with the previous prediction confidence. We set different levels of judgment for different levels of moving distance D_i and the IoT_i :

$$\mathbf{M}_i = \underbrace{(\text{IoT}_i \in [\sigma_{\text{IoT}}^{\text{low}}, \sigma_{\text{IoT}}^{\text{high}}] \wedge \mathbf{P}_i < \sigma_p)}_{\text{IoU} < \sigma_{\text{IoU}2} \wedge D_i > \sigma_{d2}} \vee \underbrace{(\text{IoT}_i < \sigma_{\text{IoT}} \wedge \mathbf{P}_i < \sigma_p)}_{\text{IoU} < \sigma_{\text{IoU}3} \wedge D_i > \sigma_{d3}} \quad (3)$$

When the predicted probability P_i and the IoU with adjacent frames are both low, a relatively high value of IoT may indicate the instrument has quickly returned to its typical location. In such cases, although the category prediction may be incorrect, the detected bounding box remains consistent with historical positions. This scenario serves as a representative example where $\mathbf{M}_i$ is True, prompting us to retain the detection without the category label for second round matching. Otherwise, we consider it a successful match in the first round, and further address the ambiguity of theta by considering following conditions: 1) For the rectangular boxes (i.e., $w/h > \sigma_{\text{rect}}$ being a certain threshold), there is a rotation mutation at $\theta = \pm 90^\circ$: We choose the candidate $\theta_c = \theta + \pi$ for $|\theta + \frac{\pi}{2}| < \sigma_\theta$, and $\theta_c = \theta - \pi$ for $|\theta - \frac{\pi}{2}| < \sigma_\theta$. 2) For the square boxes, there also exists a $\frac{\pi}{2}$ -rotational ambiguity due to the uncertainty of the long edge: Here we pick the candidate $\theta_c = \theta - \frac{\pi}{2}$ for $\theta > 0$, and $\theta_c = \theta + \frac{\pi}{2}$ for $\theta \leq 0$. Then we select the nearest angle $\theta^s \rightarrow \theta_r$ from the raw estimation θ_r and the candidate θ_c based on the angle $\hat{\theta}_{k|k-1}$ in the prior prediction $\hat{\mathbf{x}}_{k|k-1}$ for the subsequent Kalman filter update:

$$\theta^s = \underset{\theta \in \{\theta_r, \theta_c\}}{\text{argmin}} \|\hat{\theta}_{k|k-1} - \theta\| \quad (4)$$

As for the second round matching process, we employ the Hungarian matching algorithm based on a cost function with a weighted summation of IoU and IoT between the remaining

trackers and unmatched detection. The matched boxes are further used to update the Kalman filter after being processed by Equ. 4. For unmatched trackers, we continuously monitor their mismatch count and if the count exceeds a threshold, we delete the corresponding instrument tracker, which indicates that the instrument needs to be reinitialized to resume tracking when it reappears in the FoV. We further added a median local area filter to filter out the small amplitude local vibrations.

We establish a specific category tracker for the upper-left ul index of the instrument i . The matching status is determined by comparing the current detected index with the previous history result ul_{t-1}^i . The core idea is to continuously monitor the matching status of ul . If a mismatch is detected, the relevant upper-left tracker will initialize a switching process, and the new index ul_{new} will be continuously counted. When the count exceeds a certain threshold σ_{ul} , ul will be updated to ul_{new} , otherwise, it will remain unchanged, and the count will be reset to zero.

$$ul_t^i = \begin{cases} ul_{\text{temp}}, & \text{if } \sum \mathbb{1}(ul^{\text{temp}}) > \sigma_{ul} \\ ul_{t-1}^i, & \text{others} \end{cases} \quad (5)$$

To achieve stable and efficient depth computation of the instrument, we propose an adaptive stereo matching method. Specifically, we restrict the search region to the vicinity of the box to save the unnecessary matching process. In addition, we adjust the search region and the box template to a fixed size to achieve a relatively constant computational cost, as it can prevent unstable running speed caused by the size fluctuations of the instrument on the image.

Hybrid laparoscope control

The laparoscope motion employs two paradigms in a human-in-the-loop approach: fully automatic active tracking and GUI button-triggered control modes. In automatic control, the laparoscope is controlled to track the main instrument and maintain it within a designated center region, which eliminates the error between the instrument position and the generated optimal view box. The process ends when the instrument reaches the boundary line of the critical area $[\sigma_{\text{min}}^l, \sigma_{\text{max}}^l]$. To avoid the high-frequency instrument vibrations around the boundary line, an additional internal interval $[\sigma_{\text{min}}^s, \sigma_{\text{max}}^s]$ is created. Once the laparoscope control process is triggered, the goal is to return the instrument to the internal area and then to re-detect whether a new round of control is triggered in the external area. This can be used to control the instrument position in the image space and

the corresponding depth. Regarding the appropriate depth interval setting for the interior and exterior, we maintain a constant intermediate depth value with a linear scale factor of $\alpha = 0.5$ between interval lengths. Based on the above tracked instrument image position p_t and its depth d_{tool} , we first obtain the image error and depth error $e = [e_p^x, e_p^y, e_d^z]^T \in \mathbb{R}^3$ with Equ. S1, S2, and then calculate the corresponding laparoscopic camera control velocity ${}^c v_c \in \mathbb{R}^3$ in the camera frame to achieve $e \rightarrow 0$:

$${}^c v_c = J_{img}^+(p_t, d_{tool}) \cdot e + [0 \ 0 \ -k_z \cdot e_d^z]^T \quad (6)$$

where $J_{img} \in \mathbb{R}^{3 \times 3}$ is the image Jacobian matrix mapping from image velocity to laparoscopic velocity, k_z is the positive gain to control the depth error e_d^z .

For the GUI button-triggered control, we set the triggered action to adjust the 3D position of the laparoscope directly. Specifically, *zoom-in/out* action corresponds to moving the laparoscope to a forward/backward position along the z -axis ${}^b z_c$ with a distance of l_{fix} . The corresponding camera velocity is calculated as follows:

$${}^c v_c = ZVF({}^b T_c) \cdot {}^b R_c^{-1} \cdot dir \cdot l_{fix} \cdot {}^b z_c \quad (7)$$

where $dir = \{1, -1\}$ represents the zoom in/out direction respectively, ZVF is defined in Equ. 13 as z -axis virtual fixture for safety consideration. Here, an additional initialization step is required after inserting the laparoscope through the trocar to set up the virtual fixture along the z -axis. Note that the corresponding appropriate depth range for further automatic control process containing the internal depth and the interval length is also modified linearly as the insertion depth for non-sensitive adjusting effect, while the central image area remains unchanged. As for the *global* action, we calculate the laparoscope velocity ${}^c v_c$ towards the pre-set global position ${}^b p_c^{init} \in \mathbb{R}^3$ in the camera frame based on the current laparoscope position ${}^b p_c(t) \in \mathbb{R}^3$:

$${}^c v_c = {}^b R_c^{-1} \cdot ({}^b p_c^{init} - {}^b p_c(t)) \quad (8)$$

To achieve the effect of *GLOBAL-HOLD* control after reaching the global position, we disable the z -depth control by setting the velocity component $v_z = 0$ calculated in Equ. 6, while the xy motion in the 2D image plane remains unchanged.

Low-level control

Laparoscopic Manipulator Control: The motion primitives ${}^c v_c$ from automatic or GUI button-triggered mode were processed by a laparoscopic controller to obtain the desired control velocity in joint space. The controller should address two critical considerations, including the RCM constraint and misorientation elimination during the control process. Since our robot design does not incorporate a mechanical RCM structure, we employ the soft-RCM strategy [45], which maintains the RCM constraint through coordinated control of multiple robot joint angles q_t , as shown in Fig. 1B. For the misorientation elimination, here we introduce a simple but efficient approach, compared to our previous affine map-based approach [17], to calculate the misorientation by aligning x axis with the surgeon's nature line-of-sight (NLS). Specifically, we define x_{ref} as the orientation reference from the reference pose T_{ref} , then we can obtain the current laparoscope pose ${}^w T_c^t = [{}^w R_c^t \mid {}^w p_c^t] = FK(q_t) \cdot {}^e T_c$ through forward kinematics (FK) and the offline hand-eye calibration ${}^e T_c$. Then we calculate the corresponding intuitive pose

$T' = [R' \mid {}^w p_c^t]$ with intuitive orientation $R' = [x, y, z]$ under RCM constraint:

$$z = unit({}^w p_c^t - P_{rcm}), x = unit(z \times x_{ref}) \times z, y = z \times x \quad (9)$$

where P_{rcm} is the fixed RCM position. The misorientation θ^* is calculated as the z -axis rotation between the intuitive R' and current laparoscope orientation ${}^w R_c^t$.

To alleviate the misorientation problem while maintaining the RCM constraints, we implement a null-space controller that corrects the orientation by controlling the z -axis spinning rotation θ^* . The controller then transforms ${}^c v_c$ into $[{}^r v_r^T(t), {}^r \omega_r^T(t)]^T \in \mathbb{R}^6$:

$$\begin{bmatrix} {}^r v_r(t) \\ {}^r \omega_r(t) \end{bmatrix} = \begin{bmatrix} -K_r \cdot H_{xy} \frac{\tilde{r}s \cdot \tilde{r}c^T}{\|\tilde{c}s\|} \frac{\tilde{r}s \times \tilde{r}c \times \tilde{c}s}{\|\tilde{r}s \times \tilde{r}c \times \tilde{c}s\|} \\ -K_s J_d^+(t) {}^c v_c(t) \\ - (I_4 - J_d^+(t) J_d(t)) J_e^+ \cdot [0 \ 0 \ -k_\theta \theta^*]^T \end{bmatrix} \quad (10)$$

where the points r , c , and $s \in \mathbb{R}^3$ denote the RCM point, the laparoscopic camera, and the shaft point of the laparoscope attached to the end-effector, respectively. $H_{xy} = [I_2 \ 0] \in \mathbb{R}^{2 \times 3}$ is used to select the RCM error in xy direction. K_r and K_s are the positive definite gain matrices to control the RCM error and the movements ${}^c v_c$, respectively. $J_d = [e_3 \text{skew}({}^r t_c(t))] \in \mathbb{R}^{3 \times 4}$ with z -axis basis vector $e_3 \in \mathbb{R}^3$ and translation vector ${}^r t_c(t) \in \mathbb{R}^3$ from laparoscope to RCM frame, $J_e = [0 \mid I_3] \in \mathbb{R}^{3 \times 4}$ with the identity matrix I_3 . J_*^+ represents the pseudo inverse form of matrix J_* . k_θ is the positive gain used to control the misorientation θ^* . Then we convert the velocity into the UR5 end-effector velocity $[{}^b v_e^T(t), {}^b \omega_e^T(t)]^T$ in base frame and obtain the corresponding joint velocity via differential kinematics.

Uterine Manipulator Control: As the 3-DOF uterus motions are kinematically decoupled, we can calculate the joint velocity $\dot{q}_i$ based on the control command $v_{1 \sim 3} \in \{-1, 0, 1\}$ in each axis directly:

$$\dot{q}_i(t) = \begin{cases} (-1)^{v_i} K_i, & \text{if } v_i \geq 0 \wedge \int \dot{q}_i(t) dt \in [\epsilon_{min}^i, \epsilon_{max}^i] \\ 0, & \text{others} \end{cases} \quad (11)$$

where K_i indicates the pre-set constant speed for joint i . When the angular value of the joint is out of its workspace range $[\epsilon_{min}^i, \epsilon_{max}^i]$, the control velocity is reduced to zero for safety.

Besides, in LoCC-3, we implement an automatic uterus manipulation strategy based on typical surgical workflows. During hysterectomy, dissection is primarily performed around the uterus. For example, when operating on the left side, the surgical instrument moves left, the laparoscope follows to provide a leftward view, and the uterus is moved right to expose the target tissue. Guided by this surgical paradigm, surgical video analysis, and clinical expertise, a coordinated control scheme was developed to enable autonomous collaboration between the two robots. The laparoscope automatically tracks the surgical instrument, and the uterine manipulator moves the uterus in the opposite direction of the laparoscope's orientation to optimize exposure. The uterine motion is calculated as:

$$\theta_{uterus} = k \cdot \arccos(z_t \cdot z_0) \quad (12)$$

where z_t and z_0 denote the unit vector of the laparoscope's shaft axis in the current and initial coordinate frames, k is the coefficient to control the magnitude of uterus bending angle θ_{uterus} .

Enhanced Safety Consideration

For robotic laparoscope control, in addition to maximum speed constraints, we implemented a dynamic z-axis virtual fixture (ZVF) that generates the appropriate workspace based on instrument depth perception to avoid unexpected events such as excessive laparoscope extension to damaged tissue as well as protrusion into the abdominal cavity. Given the current robot end-effector pose ${}^b\mathbf{T}_e(t)$ and the pre-calibrated hand-eye calibration matrix ${}^e\mathbf{T}_c$, we first obtain the camera pose ${}^b\mathbf{T}_c(t) = {}^b\mathbf{T}_e(t) \cdot {}^e\mathbf{T}_c$ and then calculate the insertion vector $\mathbf{rc}_t = {}^b\mathbf{T}_c(t) \cdot \mathbf{e}_4 - {}^b\mathbf{T}_c^{init} \cdot \mathbf{e}_4 \in \mathbb{R}^4$ as the RCM position keep fixed in initial pose ${}^b\mathbf{T}_c^{init}$. The ZVF is implemented as follows:

$$\text{ZVF}({}^b\mathbf{T}_c) = \underbrace{{}^b\mathbf{T}_c \cdot \mathbf{e}_4 \cdot \mathbf{rc}_t^T}_{\text{within abdominal cavity}} > 0 \wedge \underbrace{\|\mathbf{rc}_t\|}_{\text{within max. insertion}} < d_{\max} \quad (13)$$

Note that when the laparoscope cannot be inserted further for magnified viewing during the surgery, the laparoscopic workspace can be extended to a deeper depth interval as required. For the robotic uterine manipulator, we avoid accidental damage to the uterus through both mechanical joint limitations and patient-specific software constraints. To avoid unexpected damage, an emergency button was used to stop the robotic laparoscope and the uterus manipulator immediately. A remote control mechanism is also provided for handling exceptional cases.

Statistical analysis

We evaluated the GUI interaction success rate by calculating the proportion of successful trials relative to the total number of trials. The pixel x/y -axis error and the time durations for each phase are reported as mean ($\pm$ SD). For the instrument tracking evaluation, t-tests were performed to compare tracking performance, with $P < 0.01$ considered to be statistically significant.

Literature cited

- [1] R. H. Taylor, A. Menciassi, G. Fichtinger, P. Fiorini, and P. Dario, "Medical robotics and computer-integrated surgery," *Springer handbook of robotics*, pp. 1657–1684, 2016.
- [2] G. I. Barbash, "New technology and health care costs—the case of robot-assisted surgery," *The New England journal of medicine*, vol. 363, no. 8, p. 701, 2010.
- [3] J. Hernandez Sanchez, W. Amanhoud, A. Billard, and M. Bouri, "Enabling four-arm laparoscopic surgery by controlling two robotic assistants via haptic foot interfaces," *The International Journal of Robotics Research*, vol. 42, no. 7, pp. 475–503, 2023.
- [4] A. Attanasio, B. Scaglioni, E. De Momi, P. Fiorini, and P. Valdastri, "Autonomy in surgical robotics," *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 4, no. 1, pp. 651–679, 2021.
- [5] M. Azizian, M. Liu, I. Khalaji, J. Sorger, D. Oh, and S. Daimios, "The da vinci surgical system," in *Handbook of Robotic and Image-Guided Surgery*, pp. 39–55, Elsevier, 2020.
- [6] M. Li, A. Kapoor, and R. H. Taylor, "Telerobotic control by virtual fixtures for surgical applications," *Advances in Telerobotics*, pp. 381–401, 2007.
- [7] A. E. Abdelaal, P. Mathur, and S. E. Salcudean, "Robotics in vivo: A perspective on human–robot interaction in surgical robotics," *Annual review of control, robotics, and autonomous systems*, vol. 3, no. 1, pp. 221–242, 2020.

- [8] F. Zhong and Y.-H. Liu, "Integrated planning and control of robotic surgical instruments for task autonomy," *The International Journal of Robotics Research*, p. 02783649231179753, 2023.
- [9] F. Richter, J. Lu, R. K. Orosco, and M. C. Yip, "Robotic tool tracking under partially visible kinematic chain: A unified approach," *IEEE Transactions on Robotics*, vol. 38, no. 3, pp. 1653–1670, 2021.
- [10] B. Lu, B. Li, Q. Dou, and Y. Liu, "A unified monocular camera-based and pattern-free hand-to-eye calibration algorithm for surgical robots with rcm constraints," *IEEE/ASME Transactions on Mechatronics*, vol. 27, no. 6, pp. 5124–5135, 2022.
- [11] F. Zhong, B. Li, W. Chen, and Y.-H. Liu, "Robot–camera calibration in tightly constrained environment using interactive perception," *IEEE Transactions on Robotics*, 2023.
- [12] L. Van den Haak, C. Alleblas, T. Nieboer, J. Rhemrev, and F. Jansen, "Efficacy and safety of uterine manipulators in laparoscopic surgery: a review," *Archives of gynecology and obstetrics*, vol. 292, pp. 1003–1011, 2015.
- [13] B. Li, Y. Lu, W. Chen, B. Lu, F. Zhong, Q. Dou, and Y.-H. Liu, "Gmm-based heuristic decision framework for safe automated laparoscope control," *IEEE Robotics and Automation Letters*, 2024.
- [14] T. Haidegger, S. Speidel, D. Stoyanov, and R. M. Sattava, "Robot-assisted minimally invasive surgery—surgical robotics in the data age," *Proceedings of the IEEE*, vol. 110, no. 7, pp. 835–846, 2022.
- [15] G.-Z. Yang, J. Cambias, K. Cleary, E. Daimler, J. Drake, P. E. Dupont, N. Hata, P. Kazanzides, S. Martel, R. V. Patel, *et al.*, "Medical robotics—regulatory, ethical, and legal considerations for increasing levels of autonomy," *Science Robotics*, vol. 2, no. 4, p. eaam8638, 2017.
- [16] P. E. Dupont, B. J. Nelson, M. Goldfarb, B. Hannaford, A. Menciassi, M. K. O'Malley, N. Simaan, P. Valdastri, and G.-Z. Yang, "A decade retrospective of medical robotics research from 2010 to 2020," *Science robotics*, vol. 6, no. 60, p. eabi8017, 2021.
- [17] B. Li, B. Lu, Y. Lu, Q. Dou, and Y.-H. Liu, "Data-driven holistic framework for automated laparoscope optimal view control with learning-based depth perception," in *2021 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 12366–12372, IEEE, 2021.
- [18] A. Murali, S. Sen, B. Kehoe, A. Garg, S. McFarland, S. Patil, W. D. Boyd, S. Lim, P. Abbeel, and K. Goldberg, "Learning by observation for surgical subtasks: Multilateral cutting of 3d viscoelastic and 2d orthotropic tissue phantoms," in *2015 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 1202–1209, IEEE, 2015.
- [19] D. Seita, S. Krishnan, R. Fox, S. McKinley, J. Canny, and K. Goldberg, "Fast and reliable autonomous surgical debriement with cable-driven robots using a two-phase calibration procedure," in *2018 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 6651–6658, IEEE, 2018.
- [20] H. Saeidi, J. D. Opfermann, M. Kam, S. Wei, S. Léonard, M. H. Hsieh, J. U. Kang, and A. Krieger, "Autonomous robotic laparoscopic surgery for intestinal anastomosis," *Science robotics*, vol. 7, no. 62, p. eabj2908, 2022.
- [21] F. Richter, S. Shen, F. Liu, J. Huang, E. K. Funk, R. K. Orosco, and M. C. Yip, "Autonomous robotic suction to clear the surgical field for hemostasis using image-based blood flow detection," *IEEE Robotics and Automation Letters*, vol. 6, no. 2, pp. 1653–1670, 2021.

- pp. 1383–1390, 2021.
- [22] M. Yip, S. Salcudean, K. Goldberg, K. Althoefer, A. Men-
ciassi, J. D. Opfermann, A. Krieger, K. Swaminathan, C. J.
Walsh, H. Huang, *et al.*, “Artificial intelligence meets medi-
cal robotics,” *Science*, vol. 381, no. 6654, pp. 141–146, 2023.
- [23] L. Maier-Hein, M. Eisenmann, D. Sarikaya, K. März,
T. Collins, A. Malpani, J. Fallert, H. Feussner, S. Giannarou,
P. Mascagni, *et al.*, “Surgical data science—from concepts
toward clinical translation,” *Medical image analysis*, vol. 76,
p. 102306, 2022.
- [24] P. Fiorini, K. Y. Goldberg, Y. Liu, and R. H. Taylor, “Con-
cepts and trends in autonomy for robot-assisted surgery,”
Proceedings of the IEEE, vol. 110, no. 7, pp. 993–1011, 2022.
- [25] J. W. Kim, T. Z. Zhao, S. Schmidgall, A. Deguet, M. Kobi-
larov, C. Finn, and A. Krieger, “Surgical robot transformer
(srt): Imitation learning for surgical tasks,” *arXiv preprint
arXiv:2407.12998*, 2024.
- [26] B. Li, B. Lu, H. Lin, Y. Wang, F. Zhong, Q. Dou, and Y.-H.
Liu, “On the monocular 3d pose estimation for arbitrary
shaped needle in dynamic scenes: An efficient visual learn-
ing and geometry modeling approach,” *IEEE Transactions
on Medical Robotics and Bionics*, 2024.
- [27] J. Wu, T. Wang, H. Guo, Y. Wang, D. Guo, B. Lu, Z. Wang,
F. Zhong, H. Wang, J. Yuan, *et al.*, “Design, control, and
experiments of a novel robotic uterine manipulator with
the motorized 3-dof manipulation rod,” *IEEE Transactions
on Biomedical Engineering*, 2023.
- [28] M.-A. Harvey, “The size of the cervix and its relationship
with age and parity,” *Urogynaecologia*, vol. 29, no. 1, 2016.
- [29] Ultralytics, “YOLOv5: A state-of-the-art real-time object
detection system.” <https://docs.ultralytics.com>, 2021.
- [30] N. Wojke and A. Bewley, “Deep cosine metric learning for
person re-identification,” in *2018 IEEE Winter Conference on
Applications of Computer Vision (WACV)*, pp. 748–756, IEEE,
2018.
- [31] X. Du, M. Allan, S. Bodenstedt, L. Maier-Hein, S. Speidel,
A. Dore, and D. Stoyanov, “Patch-based adaptive weighting
with segmentation and scale (pawss) for visual tracking in
surgical video,” *Medical image analysis*, vol. 57, pp. 120–135,
2019.
- [32] Z. Wang, B. Lu, Y. Long, F. Zhong, T.-H. Cheung, Q. Dou,
and Y. Liu, “Autolaparo: A new dataset of integrated multi-
tasks for image-guided surgical automation in laparoscopic
hysterectomy,” in *Medical Image Computing and Computer
Assisted Intervention—MICCAI 2022: 25th International Confer-
ence, Singapore, September 18–22, 2022, Proceedings, Part VII*,
pp. 486–496, Springer, 2022.
- [33] X. Gao, Y. Jin, Y. Long, Q. Dou, and P.-A. Heng, “Trans-
svnet: Accurate phase recognition from surgical videos
via hybrid embedding aggregation transformer,” in *Med-
ical Image Computing and Computer Assisted Intervention—
MICCAI 2021: 24th International Conference, Strasbourg,
France, September 27–October 1, 2021, Proceedings, Part IV*
24, pp. 593–603, Springer, 2021.
- [34] S. Voros, G.-P. Haber, J.-F. Menudet, J.-A. Long, and P. Cin-
quin, “Viky robotic scope holder: Initial clinical experi-
ence and preliminary results using instrument tracking,”
IEEE/ASME transactions on mechatronics, vol. 15, no. 6,
pp. 879–886, 2010.
- [35] B. W. King, L. A. Reisner, A. K. Pandya, A. M. Composto,
R. D. Ellis, and M. D. Klein, “Towards an autonomous robot
for camera control during laparoscopic surgery,” *Journal of
laparoendoscopic & advanced surgical techniques*, vol. 23, no. 12,
pp. 1027–1030, 2013.
- [36] M. Wagner, A. Bihlmaier, H. G. Kenngott, P. Mietkowski,
P. M. Scheikl, S. Bodenstedt, A. Schiepe-Tiska, J. Vetter,
F. Nickel, S. Speidel, *et al.*, “A learning robot for cognitive
camera control in minimally invasive surgery,” *Surgical
Endoscopy*, vol. 35, no. 9, pp. 5365–5374, 2021.
- [37] L. Jawad, A. Singh-Chudda, A. Shankar, and A. Pandya,
“A deep learning approach to merge rule-based and human-
operated camera control for teleoperated robotic systems,”
Robotics, vol. 13, no. 3, p. 47, 2024.
- [38] W. Wang, Y. Luo, J. Wang, X. Wang, and H. Song, “Robotic-
assisted laparoscopic adjustment: A meta-analysis and re-
view,” *IEEE Transactions on Instrumentation and Measurement*,
vol. 72, pp. 1–19, 2023.
- [39] B. Li, B. Lu, Z. Wang, F. Zhong, Q. Dou, and Y.-H. Liu,
“Learning laparoscope actions via video features for proac-
tive robotic field-of-view control,” *IEEE Robotics and Au-
tomatic Letters*, vol. 7, no. 3, pp. 6653–6660, 2022.
- [40] S. Eslamian, L. A. Reisner, and A. K. Pandya, “Development
and evaluation of an autonomous camera control algorithm
on the da vinci surgical system,” *The International Journal of
Medical Robotics and Computer Assisted Surgery*, vol. 16, no. 2,
p. e2036, 2020.
- [41] A. E. Abdelaal, N. Hong, A. Avinash, D. Budihal, M. Sakr,
G. D. Hager, and S. E. Salcudean, “Orientation matters: 6-
dof autonomous camera movement for minimally invasive
surgery,” *arXiv preprint arXiv:2012.02836*, 2020.
- [42] P. Culligan, E. Gurshumov, C. Lewis, J. Priestley, J. Komar,
and C. Salamon, “Predictive validity of a training protocol
using a robotic surgery simulator,” *Urogynecology*, vol. 20,
no. 1, pp. 48–51, 2014.
- [43] D. Sarlos, L. Kots, N. Stevanovic, S. von Felten, and G. Schär,
“Robotic compared with conventional laparoscopic hys-
terectomy: a randomized controlled trial,” *Obstetrics &
Gynecology*, vol. 120, no. 3, pp. 604–611, 2012.
- [44] X. Yang and J. Yan, “Arbitrary-oriented object detection
with circular smooth label,” *European Conference on Com-
puter Vision (ECCV)*, 2020.
- [45] B. Yang, W. Chen, Z. Wang, Y. Lu, J. Mao, H. Wang, and
Y.-H. Liu, “Adaptive fov control of laparoscopes with pro-
grammable composed constraints,” *IEEE Transactions on
Medical Robotics and Bionics*, vol. 1, no. 4, pp. 206–217, 2019.

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the manuscript. J.W. contributed to hardware design of uterus
manipulator. Y.L. gave several suggestions for laparoscope con-
trol and data analysis. F.Z. contributes to the discussion on
experimental design. T.C. gave the concept and advice from a
surgeon’s perspective and conducted the in-vivo hysterectomy

1 in PWH. Q.D. supervised the learning based framework for
2 OSFA system. Y.L. directed the research, provide the primary
3 idea and funding with some key suggestions about software and
4 hardware design and revised the manuscript.

5 **Conflicts of interest**

6 The authors declare no conflict of interest from external parties.